Bicycle Route Energy in Closed Form: Two Corrections, a Descent-Recovery Offset That Transfers Across Riders, and an Energy↔Time Dual

AUTHOR
Danilo Lessa Bernardineli

AFFILIATION
Pedal Hidrográfico, São Paulo

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CONTENTS

Working paper — Pedal Hidrográfico research notes (v1.0, July 2026). Self-reported benchmarks; not peer-reviewed. Two caveats govern every accuracy figure: **(i)** both engines are conditioned on each ride's *measured* power — the numbers measure consistency of the energy accounting, not blind prediction (§10.4); **(ii)** the ε calibration is in-sample on rider 1, and its cross-rider margin over a flat constant is rider- and parameter-sensitive (§8.6). Novelty claims are corpus-bounded (§10.3). The full limitation ledger is §10.4.

Danilo Lessa Bernardineli — *Pedal Hidrográfico* (collective), São Paulo, Brazil — danilo.lessa@gmail.com

Abstract

Planning community bicycle rides needs one number up front: the *energy* of a route, in kJ. We show that the cheap closed form $E \approx \alpha \cdot x + \beta \cdot (h_+ - \epsilon \cdot h_-)$, once two systematic artifacts are corrected, reproduces measured mechanical energy $\int P \cdot \mathrm{dt}$ to a **3.6% median [95% CI 2.0–5.6] over 44 power-meter rides — statistical parity with the standard forward-dynamics simulation it approximates [Martin et al. 1998] (5.1% [3.8–7.1]; the paired per-ride difference is not significant, §8.1)** — at a fraction of the cost. The two corrections carry almost all of the error: charging climb aerodynamics at the flat reference speed (fixed by an α-split), and counting fractal sub-metre ascent noise as real lifting work [Rapaport 2011] (fixed by a ~2 m deadband, or the totals-only scalar $k_{\mathrm{smooth}} = 1 - c \cdot x/h_+$, $c \approx 3m/\mathrm{km}$). Both engines run on **shared physical constants**, so every gap between them is modelling error, not a parameter mismatch — and every accuracy figure is conditioned on each ride's measured power: consistency of the energy accounting, not blind prediction.

The under-specified term is the descent credit $\epsilon \in [0, 1]$ — how much descent potential energy is recovered rather than lost to excess drag and braking. We give it a coasting-limit closed form, $\epsilon(s) = \min(1, \alpha/(\beta \cdot s))$, drop-weighted over the profile with a calibrated −0.13 offset. Across six datasets and three riders (~1,400 scored rides, two of them independent riders' full histories tested with every constant frozen), what transfers robustly is the **energy law** (~4–7% median on every corpus) and the **offset itself** (measured gaps 0.12–0.13 on all three riders). The geometric *skill* beyond a flat constant is fragile: 37% RMS reduction in-sample; a ~35% win frozen onto a coasting rider *under the generic assumed physics, narrowing to a tie under that rider's own fitted constants*; a tie-to-failure for a fast descent-pedaller. The practical rule is simple — ϵ_{geom} on open coastable terrain, flat ε ≈ 0.20 in urban stop-go — and descent recovery is unambiguously real (ε = 0 over-predicts every corpus).

Energy has a time twin. Defining an effective flat distance $x^* = x + k_+ \cdot h_+ - k_- \cdot h_-$ makes k_- the time-image of ε, and the two are inter-derivable through the shared descent power — a linkage with no located precedent, whose degenerate coasting limit independently re-derives $\epsilon_{\mathrm{coast}}$. Tested against measured moving time, the ascent half transfers to an unseen rider (6.6% median vs 7.6% naive) while the descent bridge does not predict measured descent speed: descents are behaviour-limited, so k_-, like ε's residual, is set by *how* the rider descends, not by geometry.

Deployment adds a final, structural finding: the law's *behavioural* constants are **scale-dependent**. On the deployed 5 m survey DEM the frozen law over-charges energy by +3.6 to +9.4 pp relative to a 30 m regime (922 rides); re-fitting

the constant trio (k_{smooth}, the ε offset, the climb threshold) as a pure resolution transfer — never touching measured energy — bridges that gap on the terrain regime it was fitted on, with each constant moving exactly as its mechanism predicts, but not across regimes: the constants are functions of *(sampling interval, terrain)*, not universals (§8.9). With a light σ = 10 m raster pre-smoothing plus per-rider effective constants calibrated on each rider's own history, the deployed pipeline meets a pre-registered **±5% error / ±2% bias** goal on held-out rides for all three riders — the calibration, not the smoothing, is the lever. Both engines and the shared law are deployed in three open, local-first tools (sampasimu, amora, quilojaules).

1. Introduction

A self-organised cycling collective in São Paulo plans rides by *following the city's buried hydrography — "seguir as águas"* — tracing the creeks and relief the city paved over. Planning these community rides needs one number up front: the *energy* of a route, in kilojoules — and it is worth being precise about what that number is *for*. First, energy converts to **duration**. The collective aims to be accessible to every rider power profile, so planning defaults to a deliberately low reference output (≈ 50 W average): with the reference power fixed, time is the degree of freedom, and duration $\approx E/\overline{P}$ turns a route's kJ into an arrival estimate (§5 sharpens this into the effective-flat-distance time model). Rides must be bounded in time — people have trains to catch — so computing the energy of novel, unridden routes is what makes their duration predictable at all. Second, energy is deliberately **not a difficulty score**: how easy or punishing a ride *feels* is intensity × duration, which varies per rider; energy states the route's demand on the body — a reference point, not a claim about how challenging it is. The constraint is practical: the tooling is open-data and local-first, built to run in a browser or on a self-hosted box with no build step and no cloud lock-in, which rules out heavy per-route computation as the default planning primitive.

There are two ways to put a kJ number on a route. The **canonical** way is a forward-dynamics simulation: integrate the longitudinal force balance [Martin et al. 1998],

$$m \cdot \mathrm{dv/ds} = k_{\text{eff}} \cdot P/v - C_{\text{rr}} \cdot m \cdot \mathrm{g} \cdot \cos\theta - \frac{1}{2} \cdot \rho \cdot C_{\text{dA}} \cdot (v + \text{wind})^2 - m \cdot \mathrm{g} \cdot \sin\theta,$$

regime by regime, with a brake/safe-speed cap on descents. It is accurate (here, 5.1% median absolute error against measured $\int P \cdot \mathrm{dt}$ over 44 rides) but stiff, opaque, and needs a speed solver per segment — poorly suited to interactive planning over many candidate routes or to per-edge field computation over a DEM. The **approximate** way is a closed-form steady-speed energy integral,

$$E \approx \alpha \cdot x + \beta \cdot (h_+ - \epsilon \cdot h_-),$$
$$\alpha = \left(C_{\text{rr}} \cdot m \cdot \mathrm{g} + \frac{1}{2} \cdot \rho \cdot C_{\text{dA}} \cdot (v_f + \text{wind})^2\right)/k_{\text{eff}},$$
$$\beta = m \cdot \mathrm{g}/k_{\text{eff}},$$

linear in distance x, ascent h_+ and descent h_-. Its $\alpha \cdot x + \beta \cdot h_+$ skeleton is a textbook result; the decisive — and under-specified — term is the descent credit $\epsilon \cdot h_-$. The recovery factor $\epsilon \in [0, 1]$ lumps the descent-specific losses that α (charged at the flat reference speed v_f) does not carry: the excess aerodynamic drag of descending faster than v_f, plus braking. In the nearest road-cycling-power and elevation-routing literature we find no *validated, route-level, closed-form* expression for such a lumped ϵ: the idle/coasting boundary appears as a per-grade steady-state speed condition

[Bigazzi & Lindsey 2019], and EV/e-bike energy routing treats descent recovery as a per-instant regeneration efficiency or a symmetric $m \cdot g \cdot \Delta h$ potential, never as a calibrated $\epsilon < 1$ folded into a closed-form route law.

This paper closes that gap and draws out a structural consequence. We give ϵ a closed form in the coasting limit ($E_{\text{legs}} = 0$), where it collapses to a function of grade alone,

$$\epsilon_{\text{coast}}(s) = \min\left(1, \alpha/(\beta \cdot s)\right), \qquad \alpha/\beta = C_{\text{rr}} + \tfrac{1}{2} \cdot \rho \cdot C_{\text{dA}} \cdot \left(v_f + w\right)^2/(m \cdot g),$$

with α/β the *flat-resistance grade* — the slope whose gravity exactly balances flat rolling-plus-aero resistance. Aggregated drop-weighted over a profile and calibrated with a near-constant −0.13 offset, this geometry-only estimate, $\epsilon \approx \text{clamp}_{01}\left(\epsilon_{\text{coast}} - 0.13\right)$, cuts the RMS error against a power-measured descent-energy-balance ϵ by 37% relative to the best flat constant on real descents (in-sample; §8.3). Crucially, we run the closed form and the simulation on the *same* physical constants $\left(m, C_{\text{rr}}, C_{\text{dA}}, \rho, k_{\text{eff}}, \text{wind}\right)$, so the residual gap between them is attributable to the *modelling simplifications, not the parameters*.

We then observe that energy has a **time twin**. Time is not E/P (degenerate on a coast), so it needs its own model; defining an *effective flat distance* $x^* = x + k_+ \cdot h_+ - k_- \cdot h_-$ and reading time off the flat speed ($t = x^*/v_f$) reproduces the same structure as the energy law term-for-term. The ascent coefficient $k_+ = v_f \cdot \beta/P_{\text{climb}}$ is clean and grade-independent — the equivalent-flat-distance idea with cycling precedent [Scarf & Grehan 2005; Scarf 2007] — while the descent coefficient k_- is a lumped, free parameter playing exactly the role ϵ plays for energy, with descent-time-credit precedent [Langmuir 1984; Tobler 1993]. Each half has precedent in isolation; what has no located precedent in the nearest corpus is the **linkage**: ϵ and k_- both encode the same hidden descent speed v_{desc} and become inter-derivable through the descent power $\bar{P}_{\text{desc}}$,

$$k_- = (1/s) \cdot \left[1 - \left(v_f/\bar{P}_{\text{desc}}\right) \cdot \left(\alpha - \epsilon \cdot \beta \cdot s\right)\right].$$

1.1 Contributions

- **A route-level closed-form descent-recovery factor ϵ, assessed against measured power.** A single lumped $\epsilon \in [0, 1]$ inside $E \approx \alpha \cdot x + \beta \cdot \left(h_+ - \epsilon \cdot h_-\right)$, with its coasting-limit closed form $\epsilon(s) = \min\left(1, \alpha/(\beta \cdot s)\right)$, drop-weighted aggregate, and calibrated −0.13 offset. No precedent for such a lumped, closed-form ϵ was located in the nearest cycling-power, elevation-routing, or EV/e-bike energy corpus.
- **Assessment against a power-measured descent-energy-balance** ϵ: a 37% RMS reduction over the best flat constant on real descents ($\bar{s}$ ≥ 3%, n = 22; in-sample, §8.3). Frozen and tested on two further independent riders (§8.6): the −0.13 offset recurs on both (gaps 0.12, 0.13), but the geometric skill beyond a flat constant is **fragile** — a ~35% win for a coasting rider under the generic assumed physics that *narrows to a tie under that rider's own fitted constants* (§8.6), and inconclusive-to-failing for a fast descent-pedaller. What transfers robustly across all three riders is the energy law and the offset; the ε geometry adds little beyond a flat constant for either independent rider.
- **An energy↔time duality** $x^* = x + k_+ \cdot h_+ - k_- \cdot h_-$ whose descent coefficient k_- is the time-twin of ϵ, made inter-derivable through the shared descent power $\bar{P}_{\text{desc}}$. Both halves have prior art individually; the *derivation of k_- from the same descent power as* ϵ is, to our knowledge, new. Tested against measured moving time on all three datasets (§8.8): the **ascent half transfers out-of-sample** (6.6% median vs 7.6% naive on the second

rider, significant, and beating a fitted-coefficient ceiling), while the **descent bridge does not predict measured descent speed** — k_- stays a free, behaviour-limited coefficient.
- **A shared-constants comparison design** that runs the closed form and a Martin-1998 forward simulation on identical physical constants, isolating modelling error from parameter error — together with a clean open reference implementation of the simulation (energy-conservative, semi-implicit, brake-capped, no KE floor).
- **A k_{smooth} correction for fractal cumulative ascent** inside the closed-form law. Because measured ascent is scale-dependent [Rapaport 2011], raw h_+ over-counts energy through sub-metre noise and short rollers; a per-segment ~2 m deadband (or the totals-only scalar $k_{\text{smooth}} = 1 - c \cdot x/h_+$, $c \approx 3m/\text{km}$) removes that part while leaving sustained climbs at full strength ($k_h = 1$).
- **Assessment on real, non-racing social and urban rides** reproducing measured $\int P \cdot \text{dt}$ to a 3.6% median (best closed-form variant) over 44 power-meter rides, to ~4–7% median over 62 urban São Paulo rides with a generic assumed rider, and to ~4–5% median over each of two further independent riders' histories (441 + 219 rides) with only the mass data-implied (§8.6) — with explicit physical-floor ($E_{\text{legs}} \geq m \cdot g \cdot h_+/k_{\text{eff}}$) and cadence data-quality filters, and a documented São Paulo negative result (urban stop-go riding makes ϵ behave as an approximate constant rather than tracking braking density).
- **A per-DEM ascent-bias table** k_{DEM} (a parameter-error result, not a headline modelling claim) quantifying how the choice of elevation source biases the closed-form law's h_+/h_- inputs.
- **A scale-dependence result, and a deployed calibration that meets a pre-registered accuracy goal.** The law's behavioural constants (k_{smooth}, the ε offset, the climb threshold) are functions of the elevation-sampling interval and terrain regime: the frozen law over-charges on a 5 m survey DEM (+3.6…+9.4 pp vs a 30 m regime, 922 rides), and a rider-independent re-fit of the trio at 5 m — fitted purely as a resolution transfer, never against measured energy — bridges the gap on the fitted terrain regime, each constant moving as its mechanism predicts (§8.9). On top of a σ = 10 m raster pre-smoothing, per-rider effective constants calibrated on half of each rider's history meet **±5% median error / ±2% bias on the held-out half, for all three riders** — and the ablations show the calibration, not the smoothing, carries the goal.
- **Deployment** of the shared law in three open, local-first tools: asymmetric energy *fields* over DEMs (sampasimu, which also ships the σ = 10 m mitigation), per-ride kJ records (amora), and per-segment kJ via the canonical twin (quilojaules).

2. Related work

Our two-engine design and its closed-form half draw on five distinct strands of prior art: the standard force-balance power model, equivalent-flat-distance time models, descent time-credits, the fractal nature of cumulative ascent, and energy-aware routing with recuperation. We organize the review by theme, naming the canonical reference for each and stating precisely what is standard and what we add. Every "no located precedent" claim below is corpus-bounded; the full caveat is stated once in §11.3.

2.1 The force-balance power model (canonical engine)

The instantaneous mechanical power required to ride a bicycle is, by now, textbook. The canonical statement is the validated force-balance model of [Martin et al. 1998], which sums rolling resistance, aerodynamic drag, gravity, a changing-kinetic-energy term, wheel-bearing friction, and a tyre/road term, and which was validated against direct power measurement on a flat taxiway to $R^2 = 0.97$ (SE 2.7 W), with a wind-tunnel drag area of CdA = 0.269 ± 0.006 m^2 for racing drops. The same balance underlies the cyclist "equation of motion" of [di Prampero et al. 1979] and is the model adopted by the modern simulation literature: [Dahmen & Saupe 2011] integrate it with `ode45` and validate the speed prediction on real rural tracks (explicitly excluding steep descents and braking); [Danek et al. 2020] estimate its parameters from power data; and [Li et al. 2025] optimize race pacing on a Martin-1998 power model with a genetic algorithm.

Our `canonical()` engine *is* this model — the distance-marching longitudinal force balance

$$m\,\frac{dv}{ds} = \frac{k_{eff}\,P}{v} - C_{rr}\,mg\,\cos\theta - \frac{1}{2}\rho\,C_dA\,(v+w)^2 - mg\,\sin\theta$$

integrated with a semi-implicit kinetic-energy update, a brake/safe-speed cap, and an enforced conservation identity $k_{\mathrm{eff}} \cdot \mathrm{legE} = \Delta\mathrm{KE} + W_{\mathrm{rr}} + W_{\mathrm{aero}} + W_{\mathrm{grav}} + W_{\mathrm{brake}}$. We claim nothing new about the physics here. What we add is operational: a clean, open, no-build reference implementation, and its use as the *control* arm of a shared-constants comparison (§4). One point stated precisely: [Martin et al. 1998] make small-angle simplifications *inside* this instantaneous-power model but never publish a route-level energy closed form; the closed form of §4–5 is our own derivation, not something the 1998 paper licenses.

2.2 Closed-form steady-speed energy and the lumped descent factor ε

Integrating the force balance at steady speed over a segment gives the energy skeleton $E \approx \alpha \cdot x + \beta \cdot h_+$, where α collects rolling-plus-aero cost per horizontal metre and $\beta = m \cdot g / k_{\mathrm{eff}}$ is the cost per metre climbed. That skeleton is standard — it is the textbook steady-speed energy integral, and we do not claim it.

What has no located precedent in the closest road-cycling-power and elevation-routing literature is the **lumped, route-level descent-recovery factor ε ∈ [0,1]** and its closed form. We write

$$E \approx \alpha\,x + \beta\,(h_+ - \epsilon\,h_-),$$

with ε a single scalar lumping the descent-specific losses (excess aero above flat speed, plus braking) that α — charged at flat speed v_f — does not carry, and we give it a geometry-only closed form in the coasting limit,

$$\epsilon_{coast}(s) \;=\; \min\,(1,\; \alpha/(\beta s)), \qquad \alpha/\beta = C_{rr} + \tfrac{1}{2}\rho C_dA\,(v_f + w)^2/(mg),$$

drop-weighted over the descent profile and corrected by a near-constant offset, $\epsilon \approx \mathrm{clamp}[0,1]\,(\epsilon_{\mathrm{coast}} - 0.13)$.

The nearest located precedent for the *idle/coasting boundary* is [Bigazzi & Lindsey 2019], whose negative-grade condition $v^2 \le \mu_1/(-\mu_3)$ zeroes tractive power on gentle descents — but they apply it to per-grade steady-state speed *choice*, never to a route-level closed-form recovery factor. The structural cousin in energy-optimal EV routing, [Ahmadi et al. 2024], uses a symmetric, path-independent gravitational potential $(M+m) \cdot g \cdot \Delta H$ — recovery is total and ε-free, not a calibrated $\epsilon < 1$. Across the EV/e-bike and operations-research routing literature, downhill recovery is always either a per-instant/per-speed-range regeneration efficiency [Yuan et al. 2024], a symmetric mgΔh potential, or per-edge negative arc costs solved numerically [Perger & Auer 2020]; none is a lumped route-level closed-form factor in `[0,1]`. (We note explicitly that ε is *not* the eccentric/concentric muscle-efficiency asymmetry of [Minetti et al. 2002]: ε is the cyclist's gravity-and-brake budget, not a physiological efficiency. We invoke Minetti only as a conceptual analogy.)

2.3 Equivalent-flat-distance and ascent time models

The idea of converting climbing into an equivalent length of flat riding is old in the route-choice and hiking literatures. [Scarf & Grehan 2005] give a cycling "equivalent distance" in which 1 m of climb costs roughly 8 m of flat; [Scarf 2007] refines the cycling Naismith rule to 1 m of ascent ≈ 7.92 m horizontal; and [Norman 2004] gives analogous uphill-running equivalences. Our effective-flat-distance time model

$$x^* = x + k_+ h_+ - k_- h_-, \qquad k_+ = v_f\, \beta / P_{climb},$$

extends — and does not invent — this equivalent-flat-distance idea for the ascent half: k_+ converts climb to flat time and is, like Naismith, grade-independent on steep climbs (on a climb almost all power goes into lifting, so $\mathrm{dt} = m \cdot g \cdot \mathrm{dh} / \left(k_{\mathrm{eff}} \cdot P_{\mathrm{climb}} \right)$ depends on vertical gain, not road length).

2.4 Descent time-credits

The descent half of x^* has its own precedent. [Langmuir 1984] corrects Naismith's rule with a descent term that *credits* gentle descents (−10 min per 300 m on slopes of 5–12°) but *penalizes* steep ones (+10 min per 300 m above 12°); [Tobler 1993] gives the hiking speed function $V = 6 \cdot e^{-3.5 \cdot |S + 0.05|}$, whose maximum is at a downgrade of −2.86°, so gentle descents are faster than flat. These are the route-level descent time-credit precedents for our lumped k_-. As a time concept, k_- is therefore not new, and we say so: Langmuir's gentle-credit/steep-penalty split is the *same asymmetry* that our ε and brake-cap encode on the energy side.

The genuinely additive piece is the **linkage**, not either half on its own. [Langmuir 1984] and [Tobler 1993] are empirical time fits never tied to an energy budget. We instead *derive* the descent time-credit k_- and the energy recovery factor ε from the *same* descent power $\bar{P}_{\mathrm{desc}}$: both encode the same hidden descent speed v_{desc}, and equating the time-side ($v_{\mathrm{desc}} = v_f / \left(1 - k_- \cdot s\right)$) and energy-side ($v_{\mathrm{desc}} = \bar{P}_{\mathrm{desc}} / \left(\alpha - \epsilon \cdot \beta \cdot s\right)$) expressions yields the single relation tying them together. To our knowledge, no prior work derives the descent time-credit from the same descent power as the recovery factor; this **energy↔time duality** (§5) is the novel contribution of the time model. We test it empirically in §8.8: the ascent half transfers to a second rider's measured times, but the descent bridge does not predict measured descent speed, so the duality stands as a structural claim more than a quantitative descent predictor.

2.5 Cumulative ascent as a fractal quantity

Total ascent h_+ is not a well-defined number until a scale is fixed: the finer the elevation sampling, the more sub-metre wiggle it accumulates. [Rapaport 2011] makes this precise, showing cumulative ascent is scale-dependent (a Mandelbrot-style fractal measurement problem) and stating the roller-momentum intuition — that a rider coasting over a short bump does not "pay" its full ascent — in words, but only as a qualitative caveat, with no deadband and no energy-law correction. We formalize that caveat inside the closed-form law (§6): a per-segment deadband (≈ 2 m) on the profile, or, with totals only, the scalar $k_{\mathrm{smooth}} = 1 - c \cdot x / h_+$ with $c \approx 0.003$ (= 3 m/km, a dimensionless "noise grade"). Folding k_{smooth} *into* the energy law to discount fractal-noise ascent without touching sustained climbs has, to our knowledge, no precedent; the canonical sim needs no such factor because it tracks kinetic energy and already pays roller momentum correctly.

2.6 Inferring hidden quantities and energy-aware routing

Two further strands frame our methods. First, recovering a hidden physical quantity by inverting an energy identity is exactly the logic of [Chung]'s "virtual elevation" method for estimating CdA from a power meter; our `epsFromFIT` (which recovers ε from a measured descent energy balance) and our k_{DEM} ascent-source corrections are Chung-adjacent — the inversion method is standard, only the inferred targets (the recovery factor; per-DEM ascent bias) are new. Second, in energy-aware electric-vehicle and e-bike routing, downhill *recuperation* is modelled explicitly: [Perger & Auer 2020] route EVs with regenerated (negative) edge energies, [Yuan et al. 2024] combine a symmetric mgΔh term with a per-instant regeneration efficiency, and the validation closest to ours on *real, non-racing* rides — [Gebhard et al. 2016], on WeBike e-bikes over OSM — predicts *battery range*, not mechanical $\int P \cdot \mathrm{dt}$. The contrast with these is the same throughout: their recovery is per-instant or symmetric and (in the OR work) solved numerically per edge, where ours is a single lumped closed-form factor validated against measured mechanical energy.

3. Two engines on shared constants

We model the mechanical energy of a ride twice, with two engines of deliberately different cost and fidelity, and run both on the **same physical constants**. The expensive engine is a forward-dynamics simulation of the standard road-cycling power balance [Martin et al. 1998]; the cheap one is a closed-form law that integrates that balance under a small set of simplifications. Because the two share every constant, the gap between them is attributable to the *modelling simplifications, not the parameters* — the central experimental control of this work.

3.1 The canonical engine: forward-dynamics simulation

The canonical engine, `canonical()`, integrates the longitudinal force balance of [Martin et al. 1998] by marching in distance s along the elevation profile:

$$m\frac{dv}{ds} = \frac{k_{eff}\,P}{v} - C_{rr}\,mg\cos\theta - \tfrac{1}{2}\rho\,C_dA\,(v+w)^2 - mg\sin\theta,$$

with the per-segment grade taken from the profile (slope = dh/dx, $\cos\theta = 1/\sqrt{1+\mathrm{slope}^2}$, $\sin\theta = \mathrm{slope}/\sqrt{1+\mathrm{slope}^2}$), and aero acting on the air speed $v + w$ (signed, as rel $\cdot$ |rel|). Pedal power is **regime-selected per segment** by local grade — $P = P_{\mathrm{climb}}$ where slope $\geq +2\%$, $P = P_{\mathrm{descent}}$ where slope $\leq -1.5\%$, else $P = P_{\mathrm{flat}}$. A safe-speed brake cap limits v to $v_{\max}$; on descents the kinetic energy in excess of the cap is dumped to a brake-work term W_{brake}. The default thresholds are $v_{\max} = 38\mathrm{km}/h$, $v_{\mathrm{start}} = 15\mathrm{km}/h$, with the integrator on a $\mathrm{dx} = 5m$ grid and adaptive sub-steps ($\mathrm{dt} \leq 0.25s$, $\mathrm{ds} \geq 0.2m$).

The kinetic-energy update is **semi-implicit**: the stiff propulsion term $k_{\mathrm{eff}} \cdot P/v$ is evaluated at the *new* speed by a safeguarded Newton iteration (with bisection bracketing) on

$$g(u) = u - \frac{A}{\sqrt{u}} - B, \qquad A = k_{eff}\, P\, ds \sqrt{m/2}\ , \quad B = KE - R\, ds.$$

This is energy-conservative at any step and **never injects energy**, so the predicted leg energy can never fall below the work actually done. The engine returns the leg energy legE $= \int P \cdot \mathrm{dt}$ (its predicted mechanical energy), elapsed time, the full speed profile, and the wheel-work breakdown $W_{\mathrm{rr}}, W_{\mathrm{aero}}, W_{\mathrm{grav}}, W_{\mathrm{brake}}, \Delta\mathrm{KE}$. These satisfy the enforced conservation identity

$$k_{eff} \cdot legE \;=\; \Delta KE + W_{rr} + W_{aero} + W_{grav} + W_{brake},$$

which holds to `1e-6` relative error and is used as the engine self-check. On a pure climb it guarantees legE $\geq mg \cdot h_+ / k_{\mathrm{eff}}$ — leg energy never less than the potential energy lifted. Critically, there is **no VMIN/KE floor**: such a floor would inject energy and break this inequality on underpowered steep climbs. The canonical engine tracks kinetic energy directly, so it pays the momentum cost of short rollers correctly and needs no elevation smoothing.

3.2 The approximate engine: closed-form law

The approximate engine, `approximate()`, evaluates a closed-form integral of the same balance. In its simplest (v1) form,

$$E \approx \alpha\, x + \beta\left(h_+ - \epsilon\, h_-\right),$$

with horizontal distance x, total ascent h_+, total descent h_-, and

$$\alpha = \frac{C_{rr}\, mg + \frac{1}{2}\rho\, C_d A\left(v_f + w\right)^2}{k_{eff}}, \qquad \beta = \frac{mg}{k_{eff}}.$$

Here α is energy per horizontal metre (rolling + aero, charged at the flat reference speed v_f), β is energy per vertical metre, and $\epsilon \in [0, 1]$ is the lumped descent-recovery factor developed in §5. A per-edge clamp $\max\left(0, \alpha \cdot \mathrm{dx} - \epsilon \cdot \beta \cdot |\mathrm{dh}|\right)$ on descent segments prevents negative segment energy. The leg energy is $E_{\mathrm{leg}} = E_{\mathrm{wheel}} / k_{\mathrm{eff}}$ (the legs supply *more* than the wheel receives; α, β above are already wheel-side quantities).

The current (v2) form refines this with three corrections, each of which removes a *systematic* bias measured against the power-meter rides:

$$E \approx \alpha_r\, x + \alpha_a\, x_{flat} + k_h\, k_{smooth}\, \beta\left(h_+ - \epsilon\, h_-\right), \qquad k_h = 1$$

(i) α split — charge aero only off the climbs. The rolling term is exact on any grade ($C_{\mathrm{rr}} mg \cos\theta \cdot s = C_{\mathrm{rr}} mg \cdot x$), but the aero term, billed at v_f over the *whole* distance, over-charges climbs where the rider actually moves far slower (aero $\propto v^2$). We split $\alpha = \alpha_r + \alpha_a$ and apply aero only over the non-climbing fraction:

$$\alpha_r = \frac{C_{rr}\, mg}{k_{eff}}, \qquad \alpha_a = \frac{\frac{1}{2}\rho\, C_d A\, v_f^2}{k_{eff}}, \qquad x_{flat} = x\left(1 - f_{climb}\right), \qquad f_{climb} = \frac{x_+}{x},$$

with x_+ the horizontal distance on climbing segments (slope $\geq$ the climb threshold). The headline runs zero climb aero (`climbAeroMode = 'zero'`; `'off'` is the full-aero baseline); a near-exact variant instead charges it at the quasi-steady climb speed $v_c \approx k_{\mathrm{eff}} \cdot P_{\mathrm{climb}} / \left(C_{\mathrm{rr}} mg \cos\theta + mg \sin\theta\right)$, capped at v_f. Descent aero is left untouched — on descents it is paid by gravity and already sits inside $(1-\epsilon)\cdot\beta\cdot h\_$; down-weighting it there would double-count. Empirically this correction cuts the median |Δ%| from 19.3% (baseline `off`) to 8.7% over the 44 power rides, beating `off` on 43/44 rides (median climb fraction 21%); the per-regime details are reported in §8.1.

(ii) k_h = 1 — gravity is paid in full on real climbs. A direct fit of the gravity coefficient on sustained climbs gives $k_h \approx 1$ (validated in §8.2): on real climbing $\beta \cdot h_+$ is correct and there is no uniform gravity discount.

(iii) k_smooth — trim only the rollers and DEM noise. What makes raw h_+ over-count E is not sustained climbing but short rollers and sub-metre elevation jitter. The right correction is therefore an *ascent-smoothing* factor $k_{\text{smooth}} \in (0,1]$ that trims those while leaving sustained climbs intact; it is developed in full in §6. k_{smooth} **applies to the approximate model only** — the canonical sim already pays the rollers' momentum through its kinetic-energy term.

3.3 The shared-constants design

Both engines read the *same* physical constants — $m, C_{\text{rr}}, C_{\text{dA}}, \rho, k_{\text{eff}}$, wind, plus the regime/grade thresholds — and the two are never allowed to diverge on any of them. This is the experimental control. If both engines used independently tuned parameters, a small gap between their predictions could be either a modelling artefact or a parameter mismatch, and the two would be inseparable. Holding the constants fixed makes the gap *unambiguously* the modelling simplification: the closed form's assumption of a single flat reference speed, its lumped descent recovery ε, and its linear treatment of h_+, set against the forward sim's explicit momentum and brake accounting.

Running two different engines on the *same* constants to isolate the modelling-simplification gap (rather than least-squares-calibrating one shared model to isolate *parameter* error, as in [Dahmen & Saupe 2011]) was not found in the nearest prior art; we frame it as an additive methodological contribution.

The comparison is anchored at a known calibration point. On flat ground the two engines coincide *iff* v_f equals the flat-equilibrium speed at the flat power, i.e. $v_f = \text{flatEqSpeed}(P_{\text{flat}})$, where $\text{flatEqSpeed}(P)$ solves the flat balance $\left(C_{\text{rr}}mg + \frac{1}{2}\rho C_{\text{dA}}(v+w)^2\right)\cdot v = k_{\text{eff}}\cdot P$ by bisection (this is what "auto v_f" sets). With the engines pinned to agree on the flat, every divergence elsewhere — most prominently the uphill aero over-charge corrected by the α-split above — is the genuine modelling story rather than a calibration accident.

4. The descent-recovery factor ε

4.1 Definition

The factor ε ∈ [0,1] is the single lumped parameter that the closed-form law carries that the canonical simulation does not. It absorbs all descent-specific losses that the rolling-and-aero term α — charged at the *flat* reference speed v_f — does not account for: the excess aerodynamic drag incurred by descending faster than v_f, plus braking. For a descent of grade s, the **local** recovery is the fraction of that descent's released potential energy β h₋ that is *not* wasted:

$$\epsilon(s) := 1 - \frac{(\text{aero excess} + \text{braking})\ \text{at the speed reached on grade } s}{mg\,h\_/k_{eff}}.$$

Equivalently, from the segment energy balance (neglecting the kinetic term, which telescopes over a rest-to-rest ride),

$$\epsilon(s) = \frac{\alpha\, dx - E_{legs}}{\beta\, h_-},$$

i.e. the leg energy a descent *saves* relative to riding the same horizontal distance dx on the flat, expressed as a fraction of the released potential energy $\beta\, h_-$.

Because ε enters the model only through the total descent credit $\varepsilon\, \beta\, H_-$ (with $H_- = \sum_i h_{-,i}$), a single per-ride ε is unambiguously fixed as the **descent-drop-weighted average** of the local ε(s):

$$\epsilon = \frac{\sum_i \epsilon(s_i)\, h_{-,i}}{\sum_i h_{-,i}} = \frac{1}{H_-} \int_{\text{descents}} \epsilon(s(x))\, |h'(x)|\, dx.$$

The weight is the **vertical drop** h_- — not horizontal distance, and not time. Summed over a ride this reduces to the directly measurable form

$$\epsilon = \frac{\alpha X_- - E_{legs,\,-}}{\beta H_-},$$

with X_-, $E_{legs,\,-}$ and H_- the horizontal distance, leg energy and drop summed over descent segments. This is the quantity recovered from a power track by `epsFromFIT()` on 30 m descent cells; critically, the α used there takes the *measured* flat ground speed (the time-weighted mean ground speed on |grade| < 1% cells), *not* the model's flat-equilibrium speed, so that a parameter mismatch (e.g. road C_rr applied to a gravel ride) cannot inflate α and misreport ε.

This lumped, route-level, closed-form ε ∈ [0,1] is the paper's first main claim; §2.2 locates it against the nearest prior art — Bigazzi & Lindsey's per-grade idle boundary, and the per-instant, symmetric, or per-edge-numerical treatments of recovery in EV/e-bike routing.

4.2 The coasting-limit closed form ε_coast(s)

The leg energy on a descent is bounded — $E_{legs} \geq 0$, hence $\epsilon \leq 1$. Setting $E_{legs} = 0$ (a pure coast) collapses ε(s) to a function of **grade alone**:

$$\epsilon_{coast}(s) = \min\left(1, \frac{\alpha\, dx}{\beta\, h_-}\right) = \min\left(1, \frac{\alpha}{\beta\, s}\right), \qquad \frac{\alpha}{\beta} = C_{rr} + \frac{\frac{1}{2}\rho C_d A\, (v_f + w)^2}{mg}.$$

The ratio α/β is the **flat-resistance grade** — the slope whose gravity exactly balances flat rolling-plus-aero resistance. The clamp at 1 is the gentle-descent case $s < \alpha/\beta$. Drop-weighted over the profile, or from totals only:

$$\epsilon_{coast} = \frac{1}{H_-} \sum_{\text{desc}} h_{-,i}\, \min\left(1, \frac{\alpha}{\beta s_i}\right) \qquad \text{or, lumped,} \qquad \epsilon_{coast} \approx \min\left(1, \frac{\alpha}{\beta\, \bar{s}}\right), \quad \bar{s} = \frac{H_-}{X_-}.$$

This is the geometry-only planning estimate `epsGeom()`: it needs no power track, only the route's grade profile and the rider constants. We test it, derive the −0.13 offset, and assess the resulting estimator in §8.3 (a 37% RMS reduction over the best flat constant on real descents), calibrating to

$$\epsilon \approx \text{clamp}_{[0,1]}(\epsilon_{coast} - 0.13)\,.$$

This inference machinery — inverting an energy identity to recover a hidden quantity from a power track — is methodologically Chung's virtual-elevation move [Chung], applied here to a new target (the recovery factor rather than CdA).

5. Energy↔time duality: x* = x + k₊h₊ − k₋h₋

5.1 Why time needs its own model

The naïve route $t = E/P$ is degenerate on a descent: both E → 0 and P → 0, so the quotient is ill-defined. Time is fundamentally $\int ds/v$ and needs a model of its own. We define an **effective flat distance** x* and read time off the flat reference speed, $t = x^*/v_f$:

$$x^* := x + k_+\,h_+ - k_-\,h_-.$$

The structure deliberately mirrors the energy law — a horizontal baseline, a "clean" ascent term, and a "lumped" descent term — and the parallel is exact.

5.2 The ascent half is clean and grade-independent

On a climb almost all power goes into lifting, $k_{eff}P_{climb} \approx mg\,v\,\sin\theta = mg\,dh/dt$, so $dt = mg\,dh/\left(k_{eff}P_{climb}\right)$ — climb time depends on **vertical gain, not road length**. Hence

$$k_+ = \frac{v_f\,mg}{k_{eff}P_{climb}} = \frac{v_f\,\beta}{P_{climb}}.$$

(A constant k_+ slightly double-counts the horizontal baseline already in x on gentle climbs; the exact coefficient is $v_f mg/\left(k_{eff}P_{climb}\right) - 1/s$, but the $1/s$ term vanishes on steep climbs.)

This ascent half is **not novel**. It is the cycling instance of the equivalent-flat-distance idea: Naismith-type rules that convert a metre of climb into a fixed number of flat metres — [Scarf & Grehan 2005] (cycling "equivalent distance", 1 m climb ≈ 8 m flat), [Scarf 2007] (1 m ascent ≈ 7.92 m horizontal), and the uphill-running equivalences of [Norman 2004]. We frame x* as *extending* that idea, not inventing it.

5.3 The descent half is lumped — the time-twin of ε

Descent time is **speed-limited**, not lift-limited: $t = x_-/v_{desc}$. Pinning it instead to the drop h₋ forces k_- to absorb the typical descent grade,

$$k_- \approx \frac{1 - v_f/v_{desc}}{\bar{s}},$$

so k_- is a **lumped parameter** — playing for time exactly the role ε plays for energy. We now test the time model against measured ride times (§8.8, an empirical leg absent from earlier drafts): the ascent half transfers, but k_- stays effectively **free and corpus-dependent** because the descent bridge below does not, empirically, pin it. The term-for-term correspondence is the structural heart of the duality:

	clean term	lumped term
energy $\alpha x + \beta h_+ - \epsilon\beta h_-$	$\beta = mg/k_{eff}$	ϵ
time $x + k_+ h_+ - k_- h_-$	$k_+ = v_f \beta / P_{climb}$	k_-

With $t = x^*/v_f$, the average power $\overline{P} = E/t$ then behaves correctly everywhere — it goes to 0 on a coasting descent (where E/P was degenerate) and recovers exactly the flat power αv_f on the flat.

The descent half, as a *time concept*, is also **not novel**: route-level descent time-credits are established in the hiking literature — [Langmuir 1984] (−10 min/300 m on gentle descents 5–12°, +10 min/300 m on steep > 12°) and [Tobler 1993] ($V = 6\,e^{-3.5|S+0.05|}$, speed peaking at −2.86°, so gentle descents are *faster* than flat). Langmuir's gentle-credit/steep-penalty split is the same asymmetry that ε and the canonical brake-cap encode on the energy side.

5.4 The duality is the novel piece: linking ε and k₋ through descent power

What has no located precedent is the **linkage**. ε (the energy-side lumped parameter) and k₋ (its time-side counterpart) are not derivable from one another in isolation, but they become so through the **descent power** $\overline{P}_{desc}$ — power being the exchange rate between energy and time. Both encode the same hidden descent speed v_{desc}.

From the **time** side, the descent's effective distance $x_-\left(1 - k_- s\right)$ must take real time x_-/v_{desc}:

$$v_{desc} = \frac{v_f}{1 - k_- s}.$$

From the **energy** side, the model descent leg-energy per horizontal metre is $\alpha - \epsilon\beta s$, and average power is energy × speed:

$$\overline{P}_{desc} = \left(\alpha - \epsilon\beta s\right) v_{desc} \;\Rightarrow\; v_{desc} = \frac{\overline{P}_{desc}}{\alpha - \epsilon\beta s}.$$

Equating the two expressions for v_{desc} gives the single bridge relation

$$\frac{v_f}{1 - k_- s} = \frac{\overline{P}_{desc}}{\alpha - \epsilon\beta s},$$

and hence, given $\overline{P}_{desc}$ and grade s, each lumped parameter in terms of the other:

$$k_- = \frac{1}{s}\left[1 - \frac{v_f}{\overline{P}_{desc}}\left(\alpha - \epsilon\beta s\right)\right], \qquad \epsilon = \frac{1}{\beta s}\left[\alpha - \frac{\overline{P}_{desc}}{v_f}\left(1 - k_- s\right)\right].$$

The degenerate case is instructive. Set $\overline{P}_{desc} = 0$ (a pure coast): the bridge forces $\alpha - \epsilon\beta s = 0$, i.e. $\epsilon = \alpha/\left(\beta s\right)$ — pinned by grade alone, independent of speed, recovering exactly the coasting-limit ε_coast of §4.2 — while v_{desc}, and hence k_-, is set entirely by the terminal coasting speed. With no power to bridge them, the two **decouple**: ε becomes purely geometric, k_- purely aerodynamic. They are inter-derivable only once the legs do measurable work on the descent.

In short: both halves of x* have precedent (§5.2, §5.3), but no prior work we located derives the descent time-credit k_- from the *same descent power* $\overline{P}_{desc}$ that fixes the recovery factor ε. The duality is the paper's second structural contribution — and we are precise about what kind of claim it is. It is a **derivation**, with one falsifiable quantitative prediction (the bridge's descent speed), which we test in §8.8 and find wanting: real descents are behaviour- and brake-limited, not equilibrium-limited, so k_- stays empirical. What survives is structural: the term-for-term correspondence organizes both models, and the degenerate coasting limit *independently re-derives* the ε_coast of §4.2 from the time side — an internal consistency check the energy derivation did not have to pass.

6. Cumulative ascent is scale-dependent: the k_{smooth} deadband

6.1 The fractal-ascent problem

Total ascent h_+ — the quantity the gravity term $\beta \cdot h_+$ charges against — is not a well-defined property of a route but a property of the route *as measured*. Cumulative ascent grows as the sampling/elevation resolution increases: every finer subdivision resolves additional undulation, so Σh_+ behaves like the length of a fractal coastline rather than a fixed integral. This scale-dependence of mountain-biking cumulative ascent is the subject of [Rapaport 2011], which also states *in words* the roller-momentum intuition — that a rider carries kinetic energy over short bumps and does not pay the full $\mathrm{mg} \cdot \Delta h$ on each — but only as a qualitative caveat, with no deadband and no energy-law correction.

The effect is large in our data. Over the 44 power rides, total engine ascent shrinks monotonically with the hysteresis threshold applied to the profile:

smoothing	Σ h_+ (km)	% of raw
raw	92.4	100%
1 m	83.3	90%
2 m	77.4	84%
3 m	73.3	79%
5 m	66.9	72%

About **20% of raw h_+ is sub-3 m jitter**, much of it the 0.2 m altitude quantization of consumer GPS/baro tracks. In energy terms this is not a rounding detail: $\beta \cdot h_+$ falls from **69 039 kJ** (raw) to **54 758 kJ** at a 3 m threshold, and that **14 282 kJ** difference is **25% of the empirical climb energy** and accounts for ≈ **93% of the closed-form model's climb over-prediction** in the baseline run.

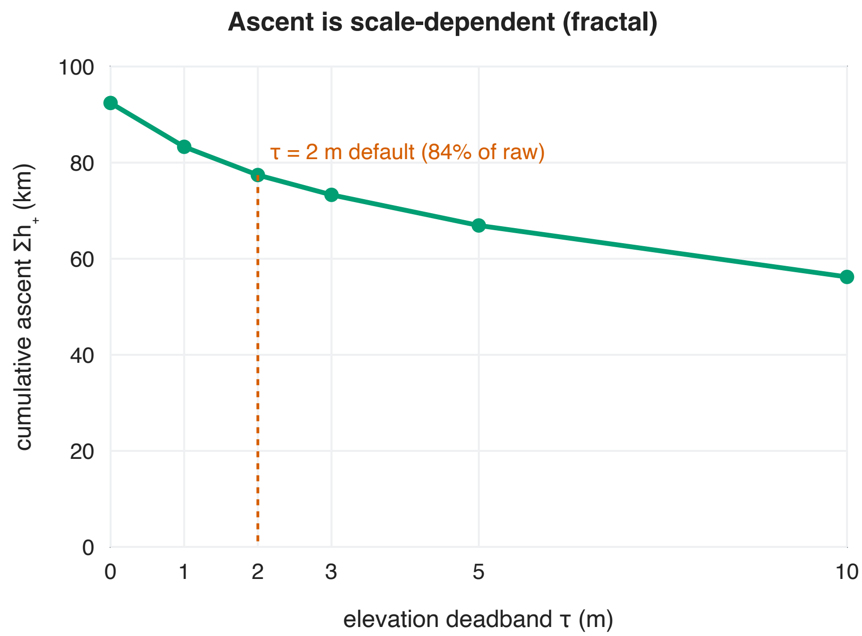

Figure 3. Cumulative ascent Σh_+ summed over the 44 rides shrinks monotonically as the deadband threshold τ rises — the coastline-paradox signature of a fractal measurement. The chosen τ = 2 m default keeps 84% of raw ascent, trimming mostly sub-metre jitter.

6.2 $k_h = 1$: the discount is in the ascent, not the gravity coefficient

A natural but wrong fix is to discount the gravity coefficient itself — to ride with some $k_h < 1$ multiplying $\beta \cdot h_+$. We reject this. Fitting k_h on **sustained** climbs only (mean slope $> 3\%$ over $> 100m$, **2535 sections** across the 44 rides) shows the rider pays the *full* gravitational cost there: measured $\int P \cdot \mathrm{dt}$ on climbs equals the expected gravity + rolling + aero to within 3% ($k_h(\text{sustained}) = 0.96$, per-ride median 1.02; full numbers in §8.2). So on real climbing $k_h \approx 1$: $\beta \cdot h_+$ is correct and there is no uniform discount. Sustained climbs are only 54% of total ascent; the other 46% is rollers, gentle grade and noise — and *that* is what raw h_+ over-counts. An earlier uniform scalar of 0.56 conflated the two effects and was an artefact. We therefore fix $k_h = 1$ and move all correction into the separate ascent-smoothing factor $k_{\text{smooth}} \in (0, 1]$:

$$E \approx \alpha_r\, x + \alpha_a\, x_{flat} + k_h\, k_{smooth}\, \beta\, \left(h_+ - \epsilon\, h_-\right), \qquad k_h = 1.$$

k_{smooth} applies to the **approximate model only**: the canonical forward simulation already tracks kinetic energy, so it pays the rollers' momentum correctly and needs no smoothing.

6.3 Two realisations of k_{smooth}

(i) The correct realisation — a per-segment deadband. A backlash/deadband filter $\text{smoothElevation}(\tau)$ with $\tau \approx 2m$ on the elevation profile keeps a 100 m climb at full strength and trims sub-τ undulation; then $h\pm$ are the smoothed sums and $k_{\text{smooth}} = 1$. As a scalar this is equivalent to

$$k_{smooth} := \frac{h_+^{\text{smoothed}}}{h_+^{\text{raw}}} \approx 0.74 \quad (2\text{ m deadband}),$$

which is also the value k_{smooth} takes for FABDEM / IGC-SP 2010 elevation.

(ii) The totals-only scalar (the "poor-man's" form). When only route totals (x, h_+) are available, the cheapest estimate exploits the fact that spurious ascent accumulates with *distance*, not terrain — a roughly constant "noise grade" the DEM or track adds:

$$h_+^{\text{corr}} = \max\left(0,\ h_+ - c\,x\right), \qquad k_{smooth} = 1 - \frac{c\,x}{h_+}, \qquad c \approx 0.003\ \left(= 3\ \text{m/km}\right).$$

Here x and h_+ are both in metres (as in $\alpha \cdot x$), so c is **dimensionless** — a ≈ 0.3% noise grade. The measured value is **3.2 m/km (IQR 2.7–3.8)**, and should be calibrated per elevation source. The form auto-adapts to terrain: $k_{\text{smooth}} \approx 0.89$ on a flat ride ($h_+/x \approx 30$ m/km) and ≈ 0.98 on a hilly one (≈ 150 m/km). The same correction is applied to h_-.

Both realisations recover energy comparably against the empirical $\int P \cdot \mathrm{dt}$; the real 2 m deadband ($k_{\text{smooth}} = 1$) is the single most accurate variant, while the totals-only scalar is unbiased but carries roughly twice the scatter — the price of having only totals rather than the profile (scoreboard in §8.1, cross-check in §8.2). We did not find $k_{\text{smooth}} = 1 - c \cdot x/h_+$ folded *inside* a closed-form energy law anywhere in the cycling-power or elevation-routing literature; [Rapaport 2011] supplies the scale-dependence diagnosis but not the correction.

One consequence of the fractal diagnosis deserves flagging here and is developed in §8.9: because h_+ is scale-dependent, *any* constant calibrated against it — k_{smooth} itself, but also the ε offset and the climb threshold — carries an implicit sampling interval. The constants in this paper are calibrated at an effective ~30 m scale; §8.9 shows that re-fitting k_{smooth} at a 5 m sampling recovers almost exactly the measured $h_+(30m)/h_+(5m)$ ratio, making the scale-dependence explicit and quantitative.

7. Validation methodology

7.1 The benchmark: measured $\int P \cdot \mathrm{dt}$

The ground truth for every ride is its **measured mechanical energy** $\int P \cdot \mathrm{dt}$, computed directly from the power-meter FIT track as a time-weighted sum $\Sigma \text{power} \cdot \mathrm{dt}$ over records. This is the quantity both engines are scored against; we do *not* use any pre-computed energy column from the source spreadsheets. (As a sanity check on the longões set, the FIT-integrated $\int P \cdot \mathrm{dt}$ agrees with the catalogue's "Work Bike" column to ≈ **0.3% median**.) All scoring is reported as the median signed and absolute percent error of each model's predicted energy relative to this benchmark, with the convention $\Delta\% = \left(\text{model} - \text{empirical}\right)/\text{empirical}$. Headline medians carry **bootstrap 95% confidence intervals** (percentile method, 10^4 resamples over rides) computed from the per-ride harness outputs, and head-to-head model comparisons on the same rides use the exact two-sided **paired sign test** — a median gap between two models is claimed only where the paired test supports it.

Feature extraction is shared across both datasets. The profile (`buildProfile()`) builds cumulative horizontal distance by haversine plus elevation at native resolution for the integrator and the h_+/h_- sums; near-duplicate points ($\Delta\text{dist} < 0.5m$) are dropped so no segment has $\mathrm{dx} = 0$, and missing elevations are linearly gap-filled. The engine grid is

$\mathrm{dx} = 5m$. Regime powers (`extractRegimePowers()`) bin each FIT record into climb/flat/descent by its grade over a **30 m distance window** (the raw per-record grade is unusable: 0.2 m altitude quantization over ~5 m/record quantizes grade to ~4%); samples below 0.5 km/h are skipped as stopped, and each regime power is the time-weighted mean. Regime thresholds throughout are climb $\geq +2\%$, descent $\leq -1.5\%$.

7.2 The six datasets

Dataset 1 — Longões (44 power rides). Of 52 catalogued rides, **44 have measured power plus a GPS track** (the remaining 8 are 6 pre-power-meter 2020 Strava rides and 2 planned routes). These are long, varied rides spanning flat, sustained-climb and real-descent terrain. For each ride the *real* `approximate()` and `canonical()` engines run on **that ride's own parameters and track**, putting closed-form approx (and its per-edge clamp), canonical `legE`, and empirical $\Sigma\mathrm{power} \cdot \mathrm{dt}$ side by side. The wiring mirrors the application's `recompute()` defaults: $\mathrm{engineDx} = 5m$, time-weighted-mean regime powers, `climbAeroMode = 'off'`, auto $v_f = \mathrm{flatEqSpeed}\left(P_{\mathrm{flat}}\right)$, $v_{\max} = 38\mathrm{km}/h$, $v_{\mathrm{start}} = 15\mathrm{km}/h$, deadband $\tau = 2m$. This dataset carries its *own* measured parameters, so it isolates the modelling gap with no assumed rider.

Dataset 2 — Censo Hidrográfico (62 clean urban rides). Short urban São Paulo social rides drawn from the censo spreadsheet's activity links (columns *Ativ. Strava* / *Ativ. RWGPS*, **RWGPS preferred**). The download funnel is **87 links → 70 downloadable** (16 are other riders' un-exportable Strava) → **69 with power → 62 after a physical-plausibility cut**. These rides are hilly but stop-go, with medians of **33 km, 454 m climb, 16.5 km/h, ~14 m·km⁻¹**. Every *factual* quantity is derived from the downloaded activity (geometry, FIT-extracted regime powers, v_f, $\int P \cdot \mathrm{dt}$); only the **rider physics is assumed**, and identically for every ride:

> $m = 78\mathrm{kg}$, $C_{\mathrm{dA}} = 0.40$, $C_{\mathrm{rr}} = 0.008$ (100% paved), $\rho = 1.13\mathrm{kg}/m^3$ (São Paulo, ~760 m, ~22 °C), $\mathrm{wind} = 0$, $k_{\mathrm{eff}} = 0.98$.

Crucially, the closed form's two calibrated constants — the ε offset −0.13 (§8.3) and the noise grade c = 3 m/km (§6.3) — are fit on Dataset 1 **only** and applied to this dataset **frozen**. The censo set is therefore not just a different riding regime: it is an out-of-sample transfer test for both constants.

For each clean ride the canonical engine is fed the ride's own climb/flat/descent powers, and two closed-form variants are compared to measured $\int P \cdot \mathrm{dt}$: a **smooth approx** on a 2 m deadband-smoothed profile, and a **poor-man's** variant on the raw profile with gravity scaled by $k_{\mathrm{smooth}} = 1 - 0.003 \cdot x/h_+$. The recovery factor ϵ is swept over the geometric ϵ_{geom} and the constant grid `{0.00, 0.10, 0.15, 0.20, 0.25}` (the underlying code also evaluates $\epsilon = 0.05$, which never sets a reported floor). One deliberate asymmetry: in the censo comparison the closed-form v_f is the model $\mathrm{flatEqSpeed}\left(P_{\mathrm{flat}}\right)$, whereas the descent-balance ϵ "truth" (`epsFromFIT`) uses the *measured* flat speed, so that a parameter mismatch cannot inflate α and misreport ϵ. The braking-mechanism analysis of §8.5 is run on the 59 of these 62 rides that also carry a usable per-ride descent-balance ε.

Dataset 3 — DEM comparison (12 rides, SP tile S24W047). To quantify how the *elevation source* biases the closed form's h_+/h_- inputs (a parameter-error, not a modelling, question), 12 rides falling inside São Paulo tile

S24W047 are sampled against five elevation sources — the recorded barometric track, the IGC-SP 2010 5 m bare-earth DTM (which covers 10 of the 12 rides and is taken as survey truth), FABDEM 30 m (bare-earth), and the COP30 and SRTM 30 m surface models — at a 3 m hysteresis with bilinear along-track sampling. Results are in §8.7.

Dataset 4 — Second rider (441 power rides, P. Paz). An **independent rider — not a member of the Pedal Hidrográfico collective** — a faster, open-road rider, shared their full Strava history export (2023-10 → 2026-07; shared with consent, held locally, never published). Of 1 054 FIT activities, **753 rides carry power**; the harness keeps rides ≥ 20 km with ≥ 99% altitude coverage, excludes 45 virtual (Zwift) rides via the FIT `file_id` manufacturer, and lands on **441 usable rides** — none excluded by the physical floor. Rider physics is assumed as in Dataset 2 (CdA 0.40, C_rr 0.008, ρ 1.13) **except the total mass, which is inverted from the rider's own sustained-climb energy balance** (the §8.2 machinery): over 10 124 sustained sections (209 km of Δh), the per-ride median implied mass is **$\hat{m}$ = 74.3 kg** [IQR 69.0–78.2]. (An independent per-activity power-balance fit later corroborated this rider's assumed physics as plausible — recovered CdA ≈ 0.26, C_rr ≈ 0.005, both in range — and confirmed the mass; journal Entry 15, §10.4.) This dataset is the paper's first **cross-rider transfer test**: every calibrated constant — the ε offset −0.13, the noise grade c = 3 m/km — is frozen from Datasets 1–2, and the rider, power meter, and riding profile (median v_f 26.6 km/h vs 16.5 urban) are all new. Results in §8.6. All datasets carry per-second timestamps, so the same FIT streams also supply measured *moving time* for the time-model test (§8.8).

Dataset 5 — Third rider (219 power rides, JAAM). A **second independent rider** — again not a collective member — shared their Strava history (2022-12 → 2026-07; with consent, held locally). Of 1 282 FIT activities, **360 carry power**, 230 ≥ 20 km; after the same filters (≥ 20 km, altitude ≥ 99%, Zwift excluded) **219 usable rides** remain, none floor-excluded. Mass is again inverted from the rider's own climbs: **$\hat{m}$ = 101.7 kg** [IQR 95.7–108.7] — high, but **rider-confirmed** (JAAM's total is ≈ 100 ± 7 kg), so the sustained-climb inversion recovered his true mass; an independent parameter estimation (journal Entry 15) also finds JAAM's CdA is a normal 0.32, ruling out an aero artifact. JAAM is a *fast* rider (median v_f 29.2 km/h). This rider's history spans several countries and terrains, but **that breadth lives almost entirely in the non-power activities**: the power rides cluster at ~737 m median altitude (the São Paulo band), so the *testable* corpus is ~93% São Paulo. JAAM serves as a second, harder cross-rider test — and, per §8.6, qualifies the first. Results in §8.6.

Dataset 6 — The author's full history (1,597 power rides, in-sample). The author's own complete Strava export (2017-08 → 2026-06; a superset of Dataset 1's longões), added *not* as a transfer test — the author is rider 1, the calibration rider — but as a large-sample validation of the machinery on the one rider whose ground truth is known. Of 2,880 FIT activities, 1,597 carry power; the standard filters leave **621 clean rides**. The sustained-climb mass inversion returns **74.5 kg** [IQR 67.6–80.8] and the independent per-activity parameter fit 71.2 kg, against the author's known ≈ 73 kg — validating the mass machinery, and resolving the earlier longões-only estimate (79.8 kg) as genuine brevet loadout rather than bias. Canonical energy lands at 6.1% median with +0.1% bias (near-zero, as expected for the calibration rider), and the −0.13 offset recurs (measured gap 0.13 on 210 real descents). Full log: journal Entry 16.

7.3 Data-quality filters: the physical floor and the cadence cross-check

The censo "physical-plausibility cut" that removes 7 rides (69 → 62) rests on a hard physical lower bound. By the canonical engine's energy-conservation identity, on a climb the measured pedalling energy must at least cover the (momentum-corrected, 2 m-deadband) climbing potential energy:

$$\int P\,dt \;\geq\; \frac{m\,g\,h_+^{\text{sim}}}{k_{eff}}.$$

Seven rides measure below this floor, down to 53% of it — physically impossible for a fully-pedalled ride, so they are excluded. (The excluded 7 also *over*-predict badly when retained, by +79 … +373%, confirming they are corrupt rather than merely recovery-rich.)

The mechanism is diagnosed, not assumed, via a **cadence cross-check** that distinguishes a power-channel dropout from a rider actually pushing the bike on foot (which would legitimately have $\int P \cdot \mathrm{dt}$ below the PE floor). For **5 of the 7** excluded rides, cadence coverage is 73–100% while the walking signal (moving < 4 km/h with cadence 0) is only ~1% — i.e. the rider *was* pedalling and the deficit is a power-sensor problem, not walking. The other two (Mirantes, 31% cadence coverage; Cânions, 56%) are ambiguous — coverage too low to rule walking out, most plausibly a fuller sensor dropout. The filter is one-sided by construction (it can only remove rides the models *over*-predict): retaining the 7 would move the canonical mean Δ% from −0.8% to +12.8%, while the medians move by only ~1 pp — so the headline medians are robust to the cut, and the means are reported on the filtered set.

8. Results

8.1 The longões scoreboard (44 power rides)

We score each model variant by its median absolute percent error against the measured mechanical energy $\int P \cdot \mathrm{dt}$, the empirical benchmark for all 44 longões rides. The sign convention throughout is $\Delta\% = \big(\text{model} - \text{empirical}\big)/\text{empirical}$; brackets are bootstrap 95% CIs on the median (§7.1). The full scoreboard, best first:

model / variant	median |Δ%|	95% CI	median Δ%
approximate `cf` **+ 2 m elevation smooth** (deadband)	**3.6**	[2.0, 5.6]	+2.2
canonical (forward sim)	5.1	[3.8, 7.1]	−1.7
canonical + 2 m elevation smooth	5.6	[4.1, 8.0]	−3.5
approximate `cf` + scalar k_{smooth} (no smoothing)	5.8	[3.5, 8.2]	−0.5
approximate `cf` + sheet v_f ($P_{\text{flat}}/P_{\text{avg}}$)	7.2	[4.3, 9.1]	−0.5
approximate `cf` + measured v_f	8.2	[4.6, 12.5]	+6.7
approximate + climb-fraction (`cf`)	8.7	[7.3, 11.2]	+8.6

model / variant	median |Δ%|	95% CI	median Δ%
approximate `off` + 2 m elevation smooth	10.2	—	+9.9
approximate `off` (baseline)	19.3	[17.5, 21.6]	+19.3

Three results stand out. First, the **closed-form approximate law, once the climb-aero over-charge is corrected (`cf`) and the profile is deadband-smoothed at τ = 2 m, reaches statistical parity with the full forward simulation** — 3.6% median |Δ%| [2.0, 5.6] against the canonical sim's 5.1% [3.8, 7.1] — at a fraction of the cost. We state the comparison precisely: the closed form wins on the median, but the CIs overlap and the paired per-ride test does not separate the two (the closed form is more accurate on 25 of 44 rides, sign test $p = 0.45$), so the defensible claim is *parity*, not superiority — which is already the practically decisive result, since the closed form is the one cheap enough to route with. Second, the corrections are not cosmetic: the raw `off` baseline (full v_f aero over the whole distance, no smoothing) sits at 19.3% and over-predicts systematically (+19.3% median Δ%), because it bills aerodynamic drag at the flat reference speed even up the climbs and counts sub-metre ascent noise as real lifting work. The climb-fraction correction alone (`cf`) halves the error to 8.7% and beats `off` on 43 of 44 rides (median climb fraction 21%); the 2 m deadband then removes the ascent-noise half.

The per-regime decomposition localizes the residual error. The canonical sim is near-exact on flat (−3.6%) and climb (+7.5%) but under-predicts descents (−17.9%, only 7% of total energy); the uncorrected `off` closed form over-predicts climbs by +48.1% — the over-charge the `cf` split and deadband target. As shown in §6.1, of raw cumulative ascent h_+ roughly 20% is sub-3 m jitter, and the corresponding 14 282 kJ of spurious $\beta \cdot h_+$ accounts for ≈ 93% of the `cf` climb over-prediction.

The conservation identity $k_{\text{eff}} \cdot \text{legE} = \Delta\text{KE} + W_{\text{rr}} + W_{\text{aero}} + W_{\text{grav}} + W_{\text{brake}}$ is machine-checked per ride (`compare.mjs`); the worst relative residual across the 44 rides is 1.8×10^{-8}, confirming the semi-implicit integrator never injects or leaks energy [Martin et al. 1998].

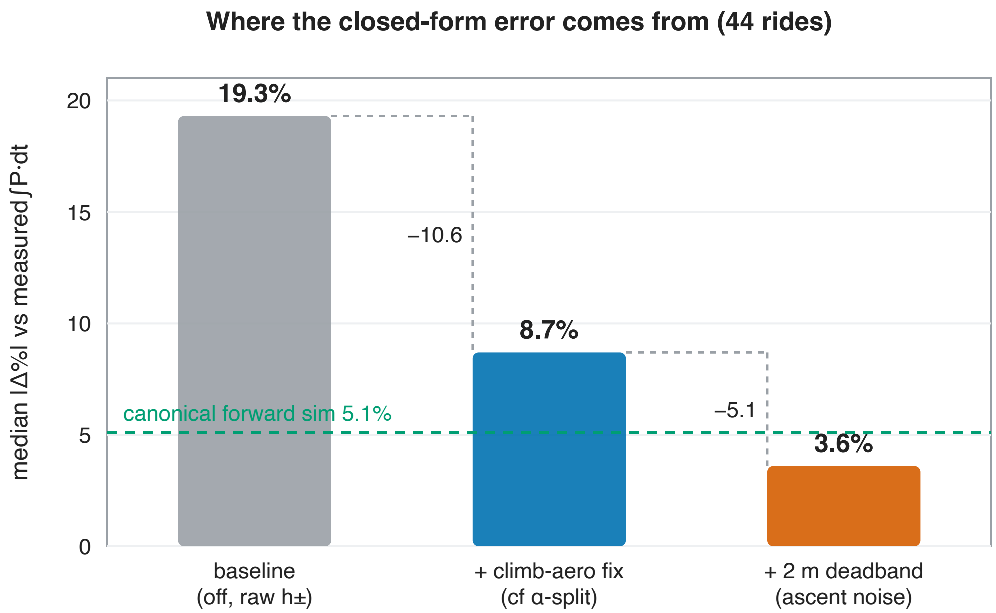

Figure 1. Where the closed form's error comes from. The raw baseline (19.3% median |Δ%|) is not diffuse model error but two identifiable artifacts: correcting the climb-aero over-charge (the `cf` α-split) removes 10.6 points, and deadbanding the fractal ascent noise (τ = 2 m) removes another 5.1 — landing at 3.6%, below the forward simulation's own 5.1% (dashed; a median-level comparison — the paired per-ride difference is not significant, see text). The full variant ranking is the table above.

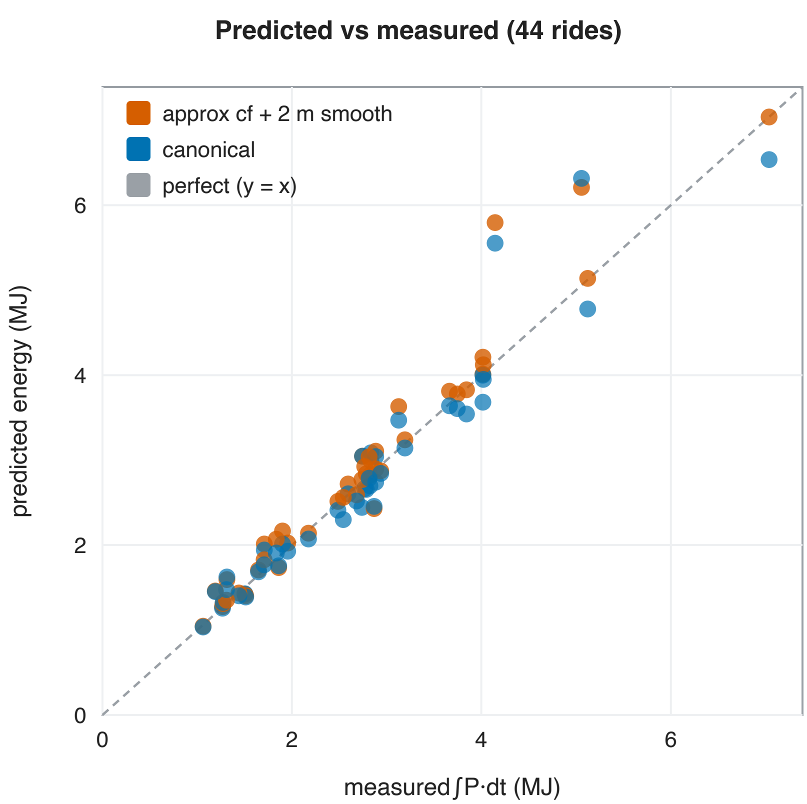

Figure 2. Predicted vs measured ride energy for the two best models, one point per ride. Both hug the identity line (dashed); the shared over-prediction on the two highest-energy rides is the same pair flagged in §8.2.

8.2 The sustained-climb fit $k_h \approx 1$ and the smoothing cross-check

The deadband smoothing is justified directly against the power meter. Fitting the gravity coefficient k_h on **sustained** climbs only (mean slope > 3% over > 100 m), across 2535 such sections over the 44 rides:

	kJ
measured Σ∫P·dt on climbs	41 790
expected (gravity 37 366 + roll 4 424 + aero 1 544)	43 333
measured / expected	**0.96**
$k_h(\text{sustained}) = (\text{measured} - \text{roll} - \text{aero})/\text{gravity}$	**0.96**

So on real, sustained climbing the rider pays essentially the full $\text{mg} \cdot \Delta h / k_{\text{eff}}$: $k_h \approx 1$ (per-ride median 1.02, range 0.57–1.23). There is no uniform gravity discount; an earlier uniform scalar of 0.56 was an artifact of mixing genuine climbing with rollers and noise (sustained climbs are only 54% of total ascent, the other 46% being rollers, gentle grade, and noise). The correct treatment is therefore to keep $k_h = 1$ (rounding 0.96 ≈ 1) and remove only the spurious ascent — either via the 2 m deadband (the "smoothed" realisation, $k_{\text{smooth}} = 1$) or via the totals-only scalar $k_{\text{smooth}} = 1 - c \cdot x/h_+$. The cross-check confirms both work, with the scalar trading bias for scatter:

model	median |Δ%|	median Δ%
smoothed (`cf` + real 2 m deadband, $k_{\text{smooth}} = 1$)	**3.6**	+2.2
canonical (forward sim)	5.1	−1.7
k_{smooth} scalar (`cf` + $1 - c \cdot x/h_+$, no smoothing)	5.8	−0.5

The scalar is essentially unbiased (−0.5%) but carries roughly twice the scatter of the explicit deadband.

8.3 The ε closed-form fit (44 power rides)

The coasting-limit closed form $\epsilon_{\text{coast}}(s) = \min(1, \alpha/(\beta \cdot s))$, with $\alpha/\beta = C_{\text{rr}} + \frac{1}{2}\rho C_{\text{dA}}(v_f + w)^2/(\text{mg})$, was tested against the per-ride descent-energy-balance ε measured from the FIT track ($\epsilon_{\text{bal}} = (\alpha \cdot X_- - E_{\text{legs},-})/(\beta \cdot H_-)$ on 30 m cells, α evaluated at the *measured* flat speed):

view	corr(ε_coast, ε_bal)	bias (ε_bal − ε_coast)
all 44 rides (unweighted)	0.30	−0.17

view	corr(ε_coast, ε_bal)	bias (ε_bal − ε_coast)
weighted by descent energy $\beta \cdot H\_$	**0.60**	−0.18
real descents, $\bar{s} \geq 3.0\%$ (n = 22)	**0.77**	−0.12
real descents, $\bar{s} \geq 3.5\%$ (n = 15)	**0.82**	−0.12

These correlations are part–whole — read with care. ϵ_{bal} and ϵ_{coast} are *not* independent: by construction $\epsilon_{\text{bal}} = \alpha/\left(\beta \cdot \bar{s}\right) - E_{\text{legs},\_}/\left(\beta \cdot H\_\right)$ and ϵ_{coast} is (a clamped, drop-weighted version of) that same first term $\alpha/\left(\beta \cdot \bar{s}\right)$, computed with the *same* per-ride α. So this is close to correlating X with $X - B$, and a shared α-error moves both together invisibly; on the s̄ ≥ 3% subset the shared geometry term $\alpha/\left(\beta \cdot \bar{s}\right)$ *alone* correlates 0.72 with ϵ_{bal} (vs the 0.77 headlined) and 0.99 with ϵ_{coast} itself — the two are nearly the same quantity. The informative statistic is therefore the **error reduction** of the calibrated estimator over a flat-constant baseline: at s̄ ≥ 3%, $\epsilon_{\text{coast}} - 0.13$ reaches RMS 0.08 against a flat-median baseline of RMS 0.13, a **37% RMS reduction** — that is the number we lead with, not the correlation. (Over *all* 44 rides the calibrated estimator actually *loses* to the flat median, skill −0.38, because of the flat-terrain reversal described next — restrict to real descents before using it.)

Figure 4. Geometry-only ϵ_{coast} vs the power-measured descent-balance ϵ_{bal}, one point per ride, point area proportional to descent energy $\beta \cdot H\_$. On real descents ($\bar{s} \geq 3\%$, vermillion) the calibrated line $\epsilon = \epsilon_{\text{coast}} - 0.13$ (green) tracks the measurement; gentle rides (grey) scatter below the identity line but carry ≈ 0 descent energy, so the miss is harmless. The part–whole caveat above applies: ϵ_{bal} and ϵ_{coast} share their dominant geometry term.

The grade law tracks ε where ε carries energy. With that caveat, correlation rises from 0.30 over all rides to 0.60 once weighted by descent energy $\beta \cdot H_{-}$, and to 0.77 / 0.82 restricting to real, coastable descents (s̄ ≥ 3.0% / 3.5%). The low all-rides correlation is harmless: gentle rides carry $\beta \cdot H_{-} \approx 0$ descent energy, so a mispredicted ε on them costs almost nothing in kJ — which is why energy-weighting lifts the correlation from 0.30 to 0.60.

Across the rides where ε matters, the residual is a near-**constant −0.13 offset** — the residual descent pedalling and braking that the pure-coasting ideal omits. (This is distinct from the unweighted −0.17 and energy-weighted −0.18 biases in the table above; the −0.13 is the offset on the real-descent rows that the estimator adopts.) Subtracting it gives the working estimator $\epsilon \approx \mathrm{clamp}_{[0,1]}\left(\epsilon_{\mathrm{coast}} - 0.13\right)$, which turns the s̄ ≥ 3% median ε_coast of 0.39 into 0.26, matching the measured 0.27, and beats the spreadsheet's flat 0.23 / 0.27 constant. A *worked example* (RMC200 Mogi): $\alpha/\beta = 0.0202$, $\bar{s} = 3.4\% \Rightarrow \min\left(1, 0.0202/0.0341\right) = 0.59$; minus 0.13 ⇒ 0.46, against a measured 0.47.

Two limits sharpen the picture. First, the clamp-to-1 prediction is **reversed on flat terrain**: gentle rides are pedalled *through* the dips, so measured ε → 0 rather than 1 (NS3 Caracaí: ε_coast ≈ 0.9, measured 0.01) — harmless, since those rides carry ≈ 0 descent energy. Second, the candidate **braking penalties do not survive** — curviness κ (rad/km) and unpaved fraction both fit with the *wrong sign* (+0.03, +0.14), confirming that the −0.13 offset, not a route-roughness term, is the right correction.

8.4 The censo ε-sweep (62 clean urban rides)

The second dataset is 62 short urban São Paulo social rides (median 33 km / 454 m climb / 16.5 km/h / ~14 m·km^{-1}), modelled with a *generic* assumed rider (m = 78 kg, CdA = 0.40, C_rr = 0.008, 100% paved) — only the rider physics is assumed; geometry, regime powers, v_f, and $\int P \cdot \mathrm{dt}$ are all derived from each activity. Sweeping ε:

model	med |Δ%|	95% CI	med Δ%	mean Δ%
canonical (fed ride powers)	6.5	[4.5, 8.5]	−3.4	−0.8
smooth approx · ε = 0.10	4.5	[3.5, 6.4]	+3.4	+5.7
smooth approx · ε = 0.15	5.0	[3.6, 5.6]	+1.3	+3.5
smooth approx · ε = 0.20	4.6	[3.4, 6.1]	−0.8	+1.2
poor-man's · ε = 0.20	**3.9**	[3.1, 6.0]	+1.1	+4.7
poor-man's · ε = 0.25	4.8	[4.0, 6.3]	−1.2	+2.1
poor-man's · ε = geom (0.29)	6.3	[4.8, 8.4]	−3.2	+1.1
smooth approx · ε = geom (0.29)	7.6	[5.8, 9.2]	−4.9	−1.9
smooth approx · ε = 0.00	7.6	[5.6, 9.5]	+7.4	+10.2
poor-man's · ε = 0.00	10.5	[8.5, 13.6]	+10.5	+15.1

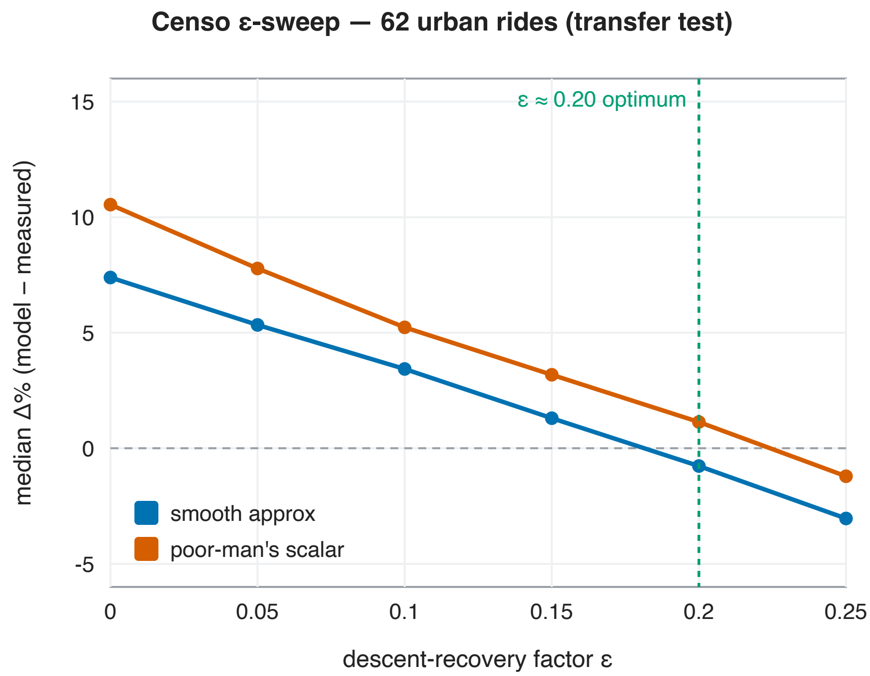

Figure 5. Median Δ% against measured energy as the recovery factor ε is swept, on the 62 urban censo rides (an out-of-sample test — both closed-form constants were fit on the longões set). Both variants cross zero near ε ≈ 0.20; ε = 0 over-predicts by +7…+11%, confirming descent recovery is real.

Three findings carry over to a completely different riding style. First, **all three models reproduce measured energy to ~4–7% median with a generic rider**, and the cheap poor-man's scalar k_{smooth} (3.9%) is as accurate as the full forward simulation (6.5%) — parity in the paired test too (the poor-man's is closer on 33 of 62 rides, sign test p = 0.70). Second, **descent recovery is physically real even in stop-go traffic**: setting ε = 0 over-predicts by +7…+11%, and the error floor sits at ε ≈ 0.15–0.20, with ε-sensitivity ~12–14 percentage points across the 0–0.29 ladder. Third, the geometric ϵ_{geom} (median 0.29) **over-credits recovery on urban stop-go riding** (it ignores the braking penalty), yielding ~3–5% under-prediction — so ϵ_{geom} is the right planning estimate on open, coastable routes, while a flat ε ≈ 0.20 fits urban stop-go.

8.5 The São Paulo negative result: ε is a constant, not braking-driven

A natural hypothesis is that the gap between the geometric prediction and the measured recovery in São Paulo's stop-go riding is set by braking density. It is not. On the 59 clean censo rides carrying a usable descent-balance ε (medians: ε_true 0.23, ε_coast 0.40, gap 0.15, sd 0.08), no candidate stop-go predictor explains the gap:

predictor for the gap (ε_coast − ε_true)	corr	R^2
Δε_brake (descent ½Δv²)	0.11	0.01
hard-brake (> 1 m/s, descent)	−0.16	0.02
all-decel ½Δv²	0.24	0.06

predictor for the gap (ε_coast − ε_true)	corr	R^2
stops/km	−0.26	0.07
v_f	0.37	0.14

v_f shows the strongest (still modest) association — plausibly because faster-descending rides simply have less braking to reconcile, not a stop-go effect — but none of these clear $R^2 \approx 0.14$, and the braking-specific predictors remain weak or wrong-signed. The mechanistic correction $\epsilon_{\text{coast}} - \Delta\epsilon_{\text{brake}}$ over-corrects: the median Δε_brake of 0.34 is roughly double the actual gap of 0.15, giving an RMS of 0.19 — *worse* than a flat constant. Ranking the estimators by RMS against ε_true:

estimator	RMS vs ε_true
flat ε = 0.20	0.08
$\epsilon_{\text{coast}} - 0.13$	**0.08**
mechanistic ($\epsilon_{\text{coast}} - \Delta\epsilon_{\text{brake}}$)	0.19
ε_coast (no penalty)	0.18

The conclusion is a **documented negative result**: the over-credit of ϵ_{coast} is ≈ 0.15, close to the open-road 0.13 of §8.3, and it does *not* track braking. The mechanism is that on a descent it is **gravity, not the legs, that repays post-stop re-acceleration**, so the legs' braking budget does not predict the recovery gap.

The estimator table also carries the paper's **out-of-sample result**, and it is worth stating plainly: the −0.13 offset was calibrated on the open longões rides (§8.3) and applied to this urban set *frozen* — and it ties the flat constant that was selected in-sample **on this very set** (RMS 0.08 vs 0.08). The calibration transfers across riding regimes without refitting. The practical rule is therefore a constant — ε ≈ 0.20 (the sweep optimum) or the transferred $\epsilon_{\text{coast}} - 0.13$ (equivalent) for urban riding, or the pure descent-balance ε ≈ 0.23 for a direct measurement (the assumed C_rr = 0.008 may still be a touch low for rough city asphalt) — and the braking correction is dropped.

8.6 The cross-rider tests: the energy law and the offset transfer; the ε skill is rider-dependent

Datasets 4 and 5 ask the question the previous sections cannot: does any of this survive a **different rider**? Both are independent riders (neither a collective member); everything below uses estimators frozen from rider 1 — nothing is refit. The two riders give *different* answers, and the contrast is the point: what transfers is the energy law and the calibrated offset; the geometric ε *skill* transfers only for riders whose descents resemble the calibration rider's.

The energy law reproduces. On the 441 clean rides, with fully assumed physics (only the mass data-implied, §7.2), all model variants land within ~5–7% median — and the ranking flips exactly as the corpus-bounded rule predicts. On this open-road corpus (median 58.2 km at v_f 26.6 km/h, ε_geom median 0.54) the **geometric ε is the best variant** — poor-man's · ε_geom reaches **4.9% median |Δ%| [95% CI 4.4–5.8] at +0.6% bias**, and this ranking is paired-significant (closer than the flat-ε = 0.20 variant on 76% of rides, and than the canonical sim on 60%, both $p < 10^{-4}$) — while the urban flat ε = 0.20 *under*-credits recovery and over-predicts by +5…+10%. The censo found the mirror image (ϵ_{geom} over-credits in stop-go; flat 0.20 wins). Together the two corpora bracket the rule from both sides: ϵ_{geom} on open, coastable riding; a flat ≈ 0.20 on urban stop-go. The canonical simulation sits at 6.8% median (+5.0% bias), consistent with slightly-off assumed drag/rolling constants for a rider we did not fit — and the independent parameter fit (Entry 15) bears this out, putting P. Paz's CdA near 0.26 against the assumed 0.40, the direction that yields a small positive energy bias.

The frozen ε estimator wins on a rider it never saw. Per-ride descent-balance ϵ_{bal} vs the geometry-only ϵ_{coast} (30 m cells, α at the measured flat speed), RMS against ϵ_{bal}:

estimator (frozen from rider 1)	all rides (n = 436)	real descents, $\bar{s} \geq 3\%$ (n = 156)
$\mathrm{clamp01}\left(\epsilon_{coast} - 0.13\right)$	**0.280**	**0.091**
flat ε = 0.20	0.484	0.227
flat ε = 0.23	0.464	0.204
in-sample flat = median ϵ_{bal}	0.356	0.139

Two things stand out. First, the calibrated geometric estimator **beats even this rider's own best flat constant by ~35%** (RMS 0.091 vs 0.139) on a real-descent subset seven times larger (n = 156) than the one it was calibrated on (n = 22), and unlike on rider 1 it also wins over *all* rides (0.280 vs 0.356) — this rider's gentle rides still coast, so the clamp's gentle-terrain reversal (§8.3) barely bites. Second, the −0.13 offset recurs: this rider's measured gap med(ε_coast) − med(ε_bal) on real descents is **0.12** (0.48 − 0.36), near the calibrated 0.13. The result is insensitive to the mass calibration — 70/74.3/78 kg moves the frozen RMS only 0.096/0.091/0.088 — and corr(ε_coast, ε_bal) = 0.81 ($\bar{s} \geq 3\%$) is strong, though part–whole (§8.3), so the frozen-vs-flat RMS is the statistic we lead with. Full log: journal Entry 12.

The 35% margin is parameter-sensitive — under this rider's own fitted physics it becomes a tie. The scoreboard above uses the generic assumed constants (CdA 0.40, C_rr 0.008), consistent with every other dataset. Rerunning with P. Paz's *own* independently fitted constants (CdA 0.26, C_rr 0.0053; journal Entry 15) lowers α, hence the measured ϵ_{bal} (0.36 → 0.14 on real descents) — and the frozen estimator's margin over his own best flat constant collapses to a tie (RMS 0.083 vs 0.086; journal Entry 16). The energy scoreboard is robust to the same swap (~5–7% median either way, with the bias flipping +5% → −7%), so this sensitivity is specific to the ε *margin*, not the law. Under each rider's best-guess physics, then, both independent riders tell the *same* story: the geometric estimator ties a flat constant. We keep the assumed-physics numbers as the headline because the whole ε framework — including the −0.13 calibration — is defined under those constants; the fitted-physics rerun is the honest error bar on the margin, and it says the 35% should not be leaned on.

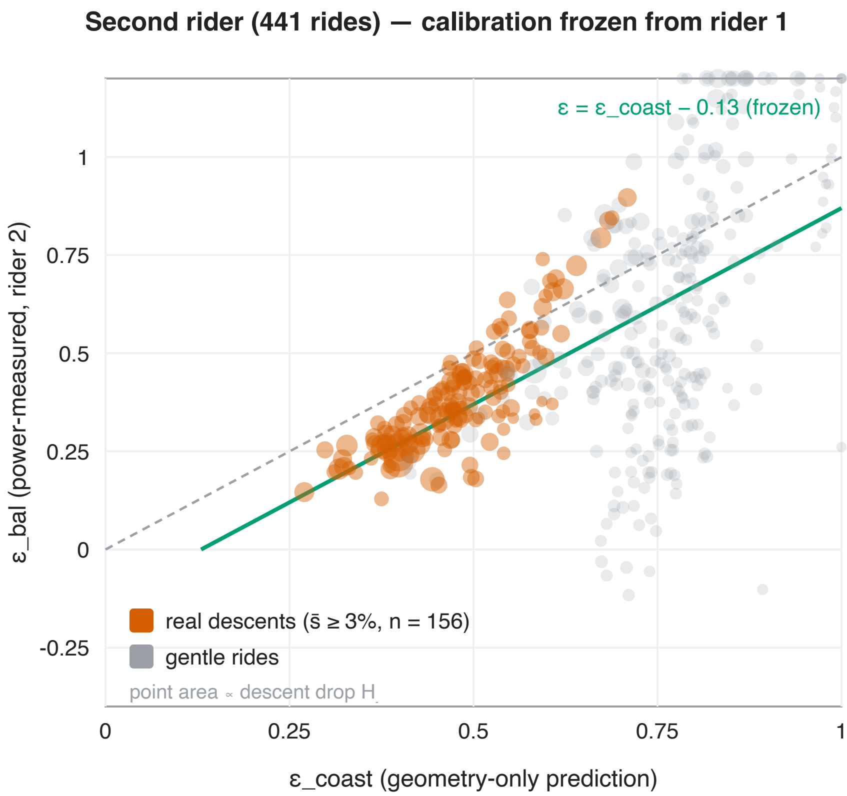

Figure 6. The second-rider test (P. Paz): geometry-only ϵ_{coast} vs power-measured ϵ_{bal} for 436 rides, with the calibration line $\epsilon = \epsilon_{coast} - 0.13$ frozen from rider 1 (green). On real descents ($\bar{s}$ ≥ 3%, vermillion, n = 156) the frozen line tracks an independent rider's measurements; point area ∝ descent energy.

The third rider (JAAM) qualifies the win — the ε skill is rider-dependent. Dataset 5 is a harder, more honest test, and it does not repeat the P. Paz result. The **energy law still transfers** — ~4–5% median error on 219 rides — but *here the flat ε ≈ 0.20 beats* ϵ_{geom} (smooth ε = 0.20 at 3.5% [95% CI 3.1–4.2], ε_geom at 5.5% [4.4–6.4]; paired, the flat constant is closer on 65% of rides, $p < 10^{-4}$), the mirror of P. Paz, because JAAM is fast (v_f 29.2 km/h) so ϵ_{geom} climbs to a median 0.61 and *over*-credits recovery. Caveat on that number: JAAM's mass is data-implied at **101.7 kg** — high, but **rider-confirmed** (≈ 100 ± 7 kg), so the sustained-climb inversion recovered his true mass rather than an artifact (an independent parameter estimation, journal Entry 15, also puts JAAM's CdA at a normal **0.32**, ruling out aero). JAAM is simply a large rider; the energy fit uses his correct mass.

The frozen ε estimator, though, **does not clearly transfer to JAAM**. On the gentle-heavy bulk it *fails outright* (RMS 0.47 vs a flat constant's 0.16): JAAM rides mostly gentle terrain (median $\bar{s}$ 1.5%) and, being strong, **pedals the descents** (measured ε_bal 0.17–0.28), so ϵ_{coast}'s coasting assumption has nothing to bite on — the §8.3 flat-terrain reversal, now at rider scale. On the thin real-descent subset ($\bar{s}$ ≥ 3%, n = 21) it is *statistically inconclusive*: frozen RMS 0.090 vs flat-0.20's 0.111 is a −0.020 difference whose bootstrap 95% CI [−0.072, 0.024] straddles zero, it merely ties JAAM's own best constant (0.085), and corr(ε_coast, ε_bal) = 0.27 is not significant (n = 21, p ≈ 0.24). The −0.13 offset does recur a third time (measured gap **0.133**, 95% CI [0.10, 0.19]) — but note that gap's *sign* is structural: ϵ_{coast} is a coasting upper bound on ϵ_{bal} (all 21 rides have ε_coast > ε_bal, the §8.3 part–whole issue), so we report it as **consistent across riders, not independently confirmed three times**.

The net across three riders and meters: **the energy law and the calibrated −0.13 offset transfer robustly; the geometric ε *skill* does not** — it wins for a coaster (P. Paz) only under the assumed physics, narrowing to a tie under his fitted constants, and is inconclusive-to-failing for a fast descent-pedaller (JAAM). Under best-guess physics, the geometric estimator adds little beyond a flat constant for either independent rider. That is the paper's standing position (§8.3: ε's residual is *rider behaviour, not route geometry*), now demonstrated across riders rather than asserted. Full logs: journal Entries 14 and 16.

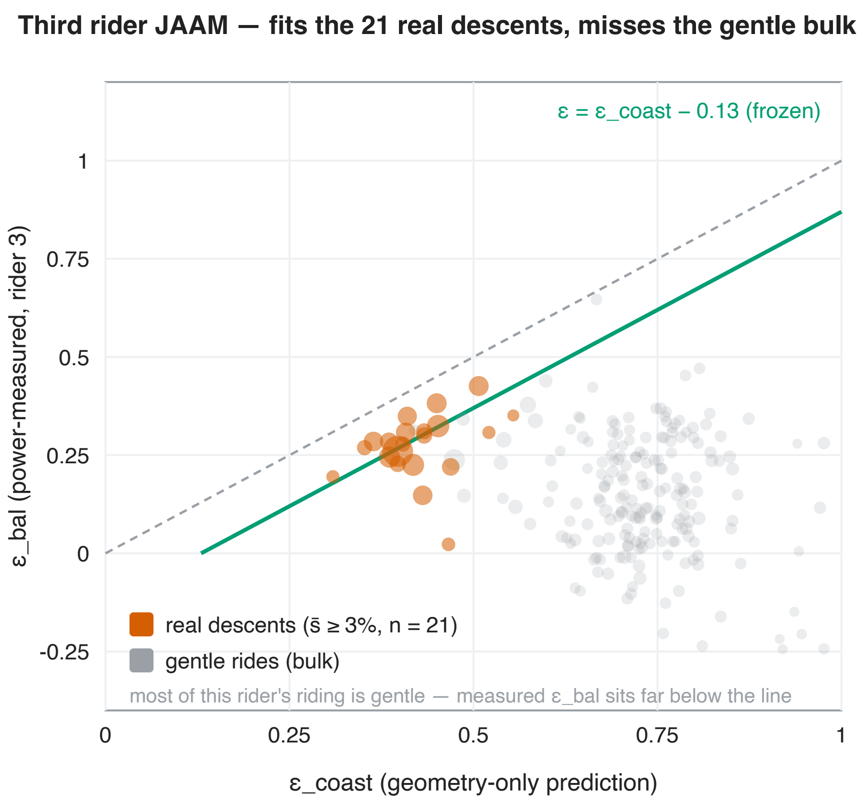

Figure 8. The third-rider test (JAAM), same axes and frozen line as Figure 6. The 21 real-descent rides (vermillion) sit near the calibration line — the estimator ties JAAM's own best constant there — but they are few and statistically inconclusive; the gentle bulk (grey, most of this fast rider's riding) sits far below the line, where ϵ_{coast} over-predicts a recovery JAAM never banks. Contrast Figure 6, where a coasting rider's real-descent cloud was dense and clearly tracked the line.

8.7 The DEM / k_{DEM} table

Because the closed-form law is linear in h_+ and h_-, the dominant source of *parameter* error in deployment is the elevation source, not the physics. Across the 12 rides in SP tile S24W047 (3 m-hysteresis, bilinear sampling), taking the bare-earth IGC-SP 2010 5 m DTM as survey truth (it covers 10 of 12 rides):

source	res	Σ h_+	vs IGC	k_{DEM}	per-ride median
recorded baro	—	13 622 (raw 15 292)	−21% (raw −11%)	1.26	1.23

source	res	Σ h₊	vs IGC	k_{DEM}	per-ride median
IGC (bare-earth)	**5 m**	**17 162**	reference	**1.00**	—
FABDEM (bare-earth)	30 m	18 160	+6%	0.95	0.93
COP30 (DSM)	30 m	20 310	+18%	0.84	0.84
SRTM (DSM)	30 m	22 951	+34%	0.75	0.72

with k_{DEM} = IGC/source. Two independent facts make this table actionable. First, **the bare-earth / surface-model distinction dominates**: the two bare-earth sources (IGC 5 m and FABDEM 30 m) agree to within 6%, while the digital *surface* models that retain canopy and buildings (COP30 +18%, SRTM +34%) systematically inflate ascent. SRTM sits ~7 m above FABDEM, and all DEMs track the recorded track shape to ~7–8 m RMS. Second, **the sampling method matters as much as the source**: nearest-neighbour sampling snaps a sub-pixel track to a staircase and adds ~30 percentage points of spurious ascent (FABDEM goes from +35% to +65%), so along-track sampling must be bilinear. The barometric track is the opposite bias — it under-records by −11% to −21% vs IGC (worst on rough/gravel, up to 1.54×) but is uniquely correct at the bridges and tunnels the DTMs cannot see. FABDEM with bilinear sampling, corrected by $k_{\mathrm{DEM}} \approx 0.95$, is therefore the practical default for São Paulo planning **on hilly terrain**, landing within 5% of 5 m survey truth there — but this table was measured on 12 rides in a hilly tile, and the agreement **does not generalize to flat lowland riding**: there FABDEM's per-pixel noise accumulates as rollers (median h_+ +57% over 864 pooled São Paulo rides, +101–135% on the flattest corpora; one 27 km flat ride reads 391 m of ascent against the survey's 99 m), inflating predicted energy by ~+18% median (§8.9; journal Entry 19). Where a validated local survey exists (IGC-SP), it is load-bearing; FABDEM remains the fallback for terrain the survey does not cover.

8.8 Testing the time model: the ascent half transfers, the descent bridge does not

Earlier drafts left the energy↔time dual of §5 as theory — no measured ride *time* had ever tested it. We close that gap here across all three datasets at once (`time_compare.mjs`; 43 longões, 58 censo, 441 P. Paz clean rides). The target is **moving time over powered segments** $T_{\mathrm{mov}} = t_+ + t_{\mathrm{flat}} + t_-$ (points with power present and v ≥ 0.5 km/h; the three regime times sum to it exactly, and cover a median 99.7% of all moving time). Stops are behaviour, not physics, and are excluded — median stopped fraction is 25% (longões), 44% (censo), 11% (P. Paz). Predicted time is $t = x^*/v_f$; we report v_f two ways, **power-conditioned** (flatEqSpeed$\left(\overline{P}_{\mathrm{flat}}\right)$, fully out-of-sample) as the headline and **speed-anchored** (measured $x_{\mathrm{flat}}/t_{\mathrm{flat}}$) only as a diagnostic, since the latter shares measured flat time with the target and is therefore partially in-sample.

Pre-declared primary endpoint (fixed before running): the full model T1b — power-conditioned v_f, $k_+ = v_f \cdot \beta/\overline{P}_{\mathrm{climb}} - 1/\overline{s}_+$, and a scalar k_- fit once on the longões and frozen — vs T_{mov} on the 441 P. Paz rides. Result: **median |Δ%| = 6.6%** [95% CI 5.9–7.2] (signed +3.8), against the naive x/v_f baseline of **7.6%** [7.0–8.5]. The gain is **modest but statistically real** — T1b beats T0 on 56% of 433 rides (sign test p = 0.011, Wilcoxon p < 0.001) — and mass-robust (6.2 / 6.6 / 7.1% at 70 / 74.3 / 78 kg). It is concentrated where the ascent term should matter: on the hilliest P. Paz tercile T0 12.0% → T1b 5.8%, while the flattest tercile is unchanged (exploratory subgroup).

predictor (power-conditioned v_f)	longões (fit)	censo (frozen)	P. Paz (frozen)
T0 naive x/v_f	16.8	20.8	7.6
Scarf literature $k_+ = 8$	8.9	14.5	8.4
T1b full (physics k_+, frozen k_-)	**5.5**	**14.2**	**6.6**
approxTime (per-segment $\int \mathrm{ds}/v$)	4.3	11.4	7.4
canonical forward sim	3.6	13.5	8.6
fitted equivalent-flat-distance ceiling	2.0	7.4	10.9

The physics k_+ transfers better than a fitted one. The fair ceiling is not a naive regression but the *same* equivalent-flat-distance model with k_+, k_- **fitted** on the longões (same per-ride v_f), then frozen. In-sample it wins (longões 2.0% vs 5.5%), because the gravity-only k_+ under-charges climb time by the rolling+aero share it omits (~26%, the time image of the §8.2 climb over-charge — an energy identity, not independent time evidence). But **frozen on the genuinely new rider the physics wins: P. Paz 6.6% vs the fitted ceiling's 10.9%** — a single fitted k_+ over-generalises across riders and speeds, where a per-ride *physical* k_+ adapts. (A naive absolute-seconds linear fit with no per-ride v_f is worse still, 26.8% frozen; that is why per-ride speed is load-bearing.) On the urban censo the fitted ceiling wins, so the physics is competitive, not dominant. With the flat speed *measured* (speed-anchored, partially in-sample) the ascent term is unambiguous — P. Paz T0 5.2% → T1b 2.0% — confirming the shape is right and the power-conditioned residual is dominated by flat-speed prediction error, not the hill terms. This is also why the fitted scalar k_- pins to 0 in power-conditioned mode: flatEqSpeed$\left(\overline{P}_{\text{flat}}\right)$ slightly over-estimates real moving-flat speed, so any descent credit only worsens the median; the speed-anchored fit ($k_- = 0.3$) shows the credit is small but real.

The descent bridge is not confirmed. The ε↔k₋ bridge predicts descent speed $v_{\text{desc}} = \overline{P}_{\text{desc}} / \left(\alpha - \epsilon \cdot \beta \cdot \bar{s}_-\right)$ (ε the frozen geometry estimator). Against measured x_-/t_- on real descents ($\bar{s}_- \geq 3\%$, $h_- \geq 50$ m, $x_- \geq 1$ km) it correlates only **0.59 / 0.08 / 0.14** (longões / censo / P. Paz) and systematically over-predicts (median measured vs predicted 30 vs 38, 16 vs 37, 32 vs 52 km/h). The analytic form is uncapped — near the $\alpha = \epsilon \cdot \beta \cdot \bar{s}$ degeneracy it diverges to unphysical speeds — and even where finite it omits the safe-speed cap the canonical engine applies: real descents are **behaviour- and cap-limited**, not aero-gravity-power-equilibrium-limited. So k_- stays a free, corpus-dependent coefficient (measured median 5.9 rural, ≈0 to negative urban, 4.8 for P. Paz), *not* pinned by the bridge. This is the exact mirror of the energy-side finding (§8.4): the geometric half of the descent model over-credits in stop-go riding because braking, not coasting physics, sets the outcome.

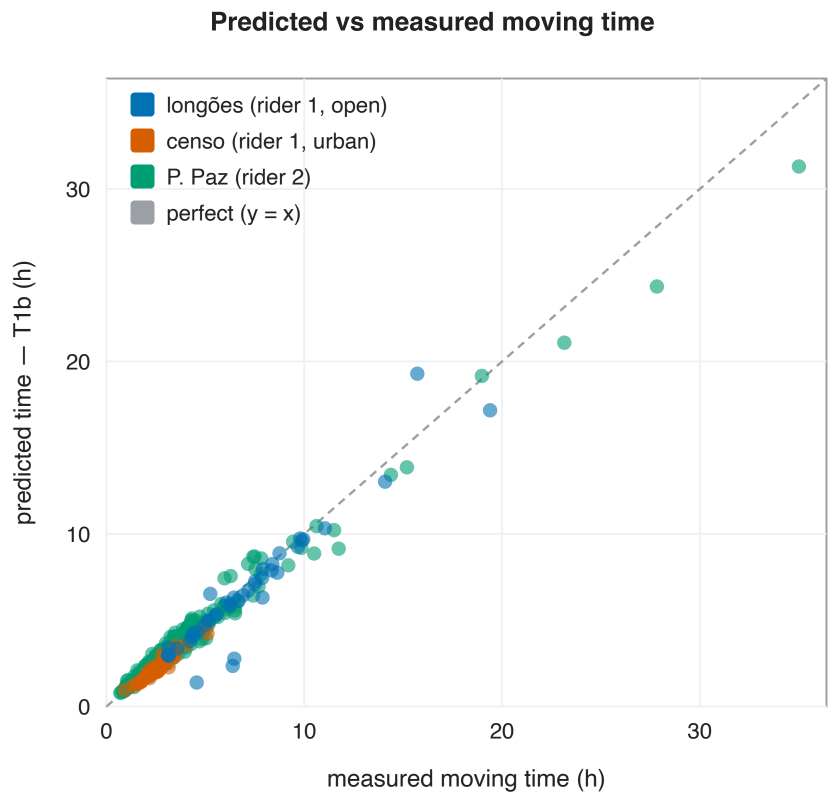

Figure 7. Predicted (T1b, power-conditioned) vs measured moving time, one point per ride, coloured by dataset, on the identity line. The ascent equivalent-flat-distance model tracks measured time across all three corpora and the full range (short urban rides to a 35-hour ultra), with small corpus-dependent bias (P. Paz +3.8%, longões −5.2%); the residual is dominated by the flat-speed prediction, not the hill terms.

Verdict — a calibrated split. The **ascent half is empirically supported and transfers** across riders (modest in aggregate, hilly-concentrated, significant, and beating a fitted ceiling on the new rider); the gravity-only climb-time law $k_+ = v_f \cdot \beta / \overline{P}_{\text{climb}}$ is the transferable piece. The **descent half is not confirmed** — the analytic ε↔k₋ bridge does not predict measured descent speed, and k_- remains an empirical, behaviour-limited coefficient. The *conceptual* duality (deriving both knobs from the shared descent power) stands as the paper's structural claim; its *quantitative* descent prediction does not.

8.9 Resolution is a parameter regime: the deployed DEM, a pre-registered accuracy goal, and scale-dependent constants

Every constant calibrated so far — the −0.13 ε offset (30 m descent cells, §8.3), the noise grade c (§6.3), the ±2%/−1.5% regime thresholds — was fitted on profiles whose effective sampling is ~30 m. Three follow-up studies (journal Entries 19–21) test the law where it actually runs — the deployed 5 m survey raster of §9.1's routing engine — and they converge on this paper's final structural finding: **the closed form's behavioural constants carry an implicit sampling scale and terrain regime.** The rider physics ($m, C_{\text{dA}}, C_{\text{rr}}, \rho, k_{\text{eff}}$) is scale-free; the behavioural trio (k_{smooth}, the ε offset, the climb threshold) is not.

(a) The deployed 5 m DEM over-charges — by resolution, not by error. Walking the per-edge law over the deployed IGC-SP-derived 5 m raster versus a 30 m average-resample of the *same* raster, on 922 São Paulo rides across all four corpora, the 5 m profiles over-charge energy by a paired median **+9.4 pp** (signed) on the urban censo rides and **+3.6 pp** pooled over the three riders' 864 rides (both $p < 10^{-4}$). Both mechanisms of §6 are measured separately: roller inflation (h_+ is higher at 5 m on 919 of 922 rides; censo median +14%) and grade-local ε collapse (the implied drop-weighted ε falls from 0.456 at 30 m to 0.414 at 5 m). The counter-intuitive part: on urban terrain the surveyed 5 m DTM is *worse* than the rider's own barometric track — real survey micro-relief that graded roads smooth away gets charged as if ridden. An aggregate-ε closed form is largely shielded (it degrades by only ~0.4–1.2 pp from 30 m to 5 m); the per-edge grade-local form is the exposed one. This is the study that disqualified FABDEM on flat lowland (§8.7) and put a pre-smoothing mitigation on the deployment roadmap.

(b) A pre-registered accuracy goal — met, and the lever is calibration, not smoothing. Can the deployed pipeline hit **median |Δ%| < 5 with |bias| < 2%**? Protocol (declared before any tuning): a deterministic 50/50 train/validation split within each rider's coverage set; two deployable levers — a mask-normalized Gaussian pre-smoothing of the 5 m raster (σ selected on train only; σ* = 10 m) and per-rider effective constants (C_{dA}, C_{rr}, k_{smooth}) fitted on the train half with mass frozen at the known value; a single frozen evaluation on validation at the end:

corpus	n (validation)	med |Δ%|	med Δ%	gate (< 5 ∧ < ±2)
P. Paz	121	**3.69**	+0.96	**PASS**
JAAM	94	**2.74**	+0.31	**PASS**
author (full)	216	**4.94**	+0.81	**PASS** (0.06 pp margin)

The ablations carry the honest decomposition. Uncalibrated, the deployed baseline fails two of three corpora (8.5 / 2.6 / 14.8); smoothing alone does not rescue it (8.7 / 2.7 / 12.0); and calibration *without* smoothing also passes all three (3.66 / 2.25 / 4.95) — **post-calibration, the smoothing buys almost nothing: the per-rider calibration is the lever that carries the goal**, with the fitted constants absorbing the resolution bias. Two caveats are part of the result: the fitted values are *effective*, not physical (one rider's C_{dA} lands at 0.21 with C_{rr} 0.014 — they soak up residual model and resolution error, and the pair is only valid with the σ it was fitted at); and the author corpus passes with 0.06 pp of margin — at-spec, not comfortably inside.

(c) The constants are scale-dependent — and terrain-dependent. The sharpest test re-fits the behavioural trio **(k_{smooth}, ε offset, climb threshold)** at 5 m as a *pure resolution transfer*: shared across riders, fitted only so that the 5 m walk reproduces the 30 m walk ride-by-ride — measured energies never enter the fit, so the trio cannot absorb rider-physics error by construction. The fit lands at $k_{smooth} = 0.94$, ε offset $= 0.063$, climb threshold $= 2.5\%$, and each constant moves exactly as its mechanism predicts: the offset roughly halves from the 30 m-calibrated 0.13 (restoring the implied drop-weighted ε to ≈ its 30 m value), and the k_{smooth}-only fit (0.884) lands almost exactly on the measured $h_+(30m)/h_+(5m)$ ratio (0.896) — the roller discount doing precisely its §6 job. So re-calibrated, the trio bridges the resolution gap on all three rider corpora, per-ride, within ~1 pp of the 30 m target. What falsifies the strong form is the transfer corpus: the flat-urban censo rides — never used in any fit, with proportionally more sub-30 m roller content (h_+ ratio 0.867 vs the riders' ~0.90) — close only about half of their gap. **The behavioural constants are therefore functions of *(sampling interval Δx, terrain-roughness regime)*, not of Δx alone**: a constant trio is an honest parameters-only bridge within one terrain regime, while per-cell raster smoothing transfers across regimes (each cell loses exactly its own sub-σ relief) — which is why smoothing, not re-fitted constants, is what ships (§9.1). And re-fitting per-rider physics *on top* of the corrected trio still yields effective, not physical, values — the residual behavioural error (drafting, position, meter) escapes into whichever constant is free.

The section inverts the paper's opening frame, and it is worth saying plainly: within its family, the closed form is already good enough — **the constants, not the model, are the accuracy frontier**, they carry an implicit *(scale, terrain)* regime, and the per-rider ones are learnable from a rider's own history (~100–200 rides suffice for the goal above).

8.10 A planner's recipe

Everything above compresses into a checklist for putting a kJ (and time) number on a candidate route, given only its geometry and a rider guess:

1. **Elevation.** Use a bare-earth source with *bilinear* along-track sampling. Prefer a validated local survey (IGC-SP 5 m); FABDEM × $k_{\mathrm{DEM}} \approx 0.95$ is the fallback and lands within ~5% of survey truth **on hilly terrain only** — on flat lowland its pixel noise reads as rollers and inflates h_+ by +57% or more (§8.7, journal Entry 19). Never nearest-neighbour-sample: it adds ~30 points of spurious ascent.
2. **Ascent.** Deadband the profile at τ ≈ 2 m and use the smoothed h_+/h_- (the most accurate variant). With route totals only, use $k_{\mathrm{smooth}} = 1 - 0.003 \cdot x/h_+$ instead — unbiased, about twice the scatter (§6.3, §8.2).
3. **Constants.** m = rider + bike + gear; CdA $\approx 0.35 - 0.40$ upright, $C_{\mathrm{rr}} \approx 0.008$ paved; $k_{\mathrm{eff}} \approx 0.98$; ρ from altitude. Set $v_f = \mathrm{flatEqSpeed}\left(P_{\mathrm{flat}}\right)$ from the rider's sustainable flat power — this is the one input that most needs to be right (§8.8).
4. **ε.** Open, coastable terrain: $\epsilon = \mathrm{clamp}_{01}\left(\epsilon_{\mathrm{coast}} - 0.13\right)$ from the grade profile (§4.2). Urban stop-go: flat $\epsilon \approx 0.20$. The margin between them is small — when in doubt, use 0.20 (§8.4–8.6).
5. **Energy.** $E \approx \alpha_r \cdot x + \alpha_a \cdot x_{\mathrm{flat}} + \beta \cdot \left(h_+ - \epsilon \cdot h_-\right)$, aero charged off-climb only (§3.2). Expect **~4–7% median** against a power meter, *conditioned on the rider-power estimate being right*.
6. **Time.** $t = x^*/v_f$ with $x^* = x + \left(v_f \cdot \beta/P_{\mathrm{climb}} - 1/\bar{s}_+\right) \cdot h_+$; leave $k_- \approx 0$ unless the flat speed is measured (§8.8). Expect ~7% median on *moving* time — then add stop time separately, because stopping is behaviour, not physics (10–45% of elapsed time depending on the setting).
7. **Calibration** (when the rider has a power-meter history). Fit effective $\left(C_{\mathrm{dA}}, C_{\mathrm{rr}}, k_{\mathrm{smooth}}\right)$ on the rider's own rides — mass frozen at the known value — and, on fine (≲ 10 m) rasters, pre-smooth with a σ = 10 m Gaussian: this validated configuration meets ±5% median error / ±2% bias on held-out rides for every rider tested, and the calibration, not the smoothing, is the main lever (§8.9). Read the fitted values as *effective* constants tied to their (σ, DEM) regime, not as physical measurements. Mind the scale generally: the §4–6 behavioural constants carry an implicit ~30 m sampling regime — at other resolutions, re-fit them or smooth to the calibration scale.

9. Applications and deployment

The same closed-form law is the shared physical core of three sibling projects in the *Pedal Hidrográfico* ecosystem — a collective that plans rides by following São Paulo's buried hydrography (*"seguir as águas"*). Each reuses the law at a different fidelity: as a per-edge routing cost, as a recorded per-ride number, or by way of its canonical twin. All three are static, build-step-free, local-first, and self-hostable.

9.1 sampasimu (Simujaules) — the closed-form law as a per-edge graph cost

sampasimu computes **asymmetric-cost cycling energy fields** over DEMs: Dijkstra on an 8-connected grid (plus A* top-N routes, multi-reference density, and layered-DP maximum-cost paths), entirely in a Web Worker, with an optional Rust+rayon backend kept at byte-level bit-parity. Every engine routes through one cost function, which is the per-edge realisation of the energy law of §3. Verbatim from `energy-worker.js`:

```
function v2Edge(dist, dh, c) {
  if (dh >= 0) {
    const aero = (dh < c.climbThr * dist) ? c.aAero * dist : 0;
    return c.aRoll * dist + aero + c.beta * dh;
  }
  const ndh = -dh;
  let eps = c.abRatio * dist / ndh;
  if (eps > 1) eps = 1;
  eps -= c.epsOffset;
  if (eps < 0) eps = 0;
  const e = c.aRoll * dist + c.aAero * dist - eps * c.beta * ndh;
  return e < 0 ? 0 : e;
}
```

The cost bundle decomposes α and β into exactly the constants of §3:

- aRoll $= m \cdot g \cdot C_{\mathrm{rr}}/k_{\mathrm{eff}}$ — kJ per ground metre, charged on all distance;
- aAero $= \frac{1}{2} \cdot \rho \cdot \mathrm{CdA} \cdot v_f^2/k_{\mathrm{eff}}$ — kJ per ground metre, charged only *off* the climbs;
- beta $= m \cdot g \cdot k_s/k_{\mathrm{eff}}$ — kJ per metre of ascent, with k_s the profile-smoothing factor (§6.2–6.3; $k_s = 1$ disables it — the per-edge engine already pays roller momentum implicitly, so smoothing is opt-in);
- abRatio $= C_{\mathrm{rr}} + \frac{1}{2}\rho\mathrm{CdA} \cdot v_f^2/\left(m \cdot g\right)$ (= α/β, deliberately computed from the **un-smoothed** coefficients even when $k_s < 1$, since ε is a grade-geometry factor, not an energy one), with epsOffset = 0.13 and climbThr ≈ 0.02.

Per directed edge (`dist` = ground length in metres, `dh` = signed rise):

- **dh ≥ 0** (uphill / flat): $\mathrm{aRoll} \cdot \mathrm{dist} + \left(\mathrm{grade} < \mathrm{climbThr} ? \mathrm{aAero} \cdot \mathrm{dist} : 0\right) + \mathrm{beta} \cdot \mathrm{dh}$;
- **dh < 0** (downhill): $\max\left(0, \mathrm{aRoll} \cdot \mathrm{dist} + \mathrm{aAero} \cdot \mathrm{dist} - \epsilon \cdot \mathrm{beta} \cdot |\mathrm{dh}|\right)$, with the geometric recovery factor $\epsilon = \mathrm{clamp}_{01}\left(\min\left(1, \mathrm{abRatio} \cdot \mathrm{dist}/|\mathrm{dh}|\right) - 0.13\right)$ computed **per edge** — a realisation choice not spelled out in `notas.md` or §4, which define the −0.13 offset on the drop-weighted *aggregate* ε (§4.1, §8.3). The two coincide exactly wherever the clamp doesn't bind, and diverge only on profiles with a substantial share of descent edges steeper than ε's floor grade (≈14%). The per-edge form is a **routing-driven realisation, not the more physical one**: a Dijkstra edge cost must be locally additive, which forces the choice — but ε is *by construction* a drop-weighted aggregate (§4.1) bundling whole-descent phenomena (excess aero, braking, descent pedalling) that a single edge cannot resolve. Two empirical facts sharpen this (journal Entry 18). First,

the trailing $\max(0, \cdot)$ is **provably dead code**: the grade-local ε keeps every descent edge's cost ≥ $0.13 \cdot \alpha \cdot d$ (the same bound as the A* admissibility floor), confirmed on ~1,400 real rides where it never fired — so no credit is ever clamped away (that failure mode belongs to a ride-frozen ε applied per edge, a different construction). Second, the real deviation from the aggregate champion is **resolution**: grade-local $\varepsilon(s)$ collapses on steep fine-grained edges, so on a 5 m grid the per-edge cost runs *above* the aggregate form, while at a ~30 m sampling the two roughly tie. That matters because the deployment's usual raster is the 5 m IGC-SP survey DTM — walked raw, it over-charges ride energy by a paired +3.6 to +9.4 pp against a 30 m regime (§8.9). sampasimu therefore now pre-smooths fine DTMs (pixel ≤ 10 m) at DEM load with the validated mask-normalized σ = 10 m Gaussian (v55; user-overridable, and heights are smoothed app-side and shipped identically to the JS and Rust engines, so bit-parity is untouched), and the per-rider effective constants of §8.9's calibration are exactly the app's parameter panel. For estimating a *ride's* energy, use the aggregate ε; per-edge is the price of edge-additivity in a routing field — best paid at a coarse effective grid, which the pre-smoothing provides.

This is the **asymmetric, downhill-clamped** realisation of $E \approx \alpha \cdot x + \beta \cdot \left(h_+ - \epsilon \cdot h_-\right)$, with the directionality (an edge is cheap downhill, expensive up) that makes the energy *field* asymmetric. The identical `v2Edge` expression — full geometric ε and climb-aero gating included, not a bare gravity term — is reused for bridge/tunnel portal edges on $(\text{deckLenM}, \pm\,\text{dh})$, with the `reverse` direction reading the opposite-direction cost — built at bit-parity between the JS and Rust engines. **Deployment:** `https://simujaules.pedalhidrografi.co`.

9.2 amora — recording per-ride kJ in RDF

amora is the flagship web-map PWA with a self-hosted Flask backend, all state in RDF/Turtle. It does *not* run the model; it **records its output**. Each tour carries `ph:energyEstimate` and `ph:measuredEnergy` as plain `xsd:decimal` kJ literals (`sh:minInclusive 0`, `sh:maxCount 1`, `sh:Warning` in `TourShape`) — the unit (kJ) is implicit in the property name, having been flattened from former `qudt:QuantityValue` nodes. A qualitative intensity label is *not* stored but derived from the kJ value in the readers (`censo.html`, `app.js`, `backend/main.py`) by fixed bands: *De boa* (0–150), *Ok* (150–300), *Endorfinado* (300–500), *Frito* (500–1000), *Insano* (≥ 1000). amora is thus the consumer of an energy number — estimated via the shared law of §3, or measured — rather than an implementer of it. **Deployment:** `https://amora.pedalhidrografi.co` (self-hosted; also runs locally).

9.3 quilojaules — the canonical twin, per-segment

quilojaules is a static per-segment kJ/calories calculator (FABDEM elevation + OSRM routing + `shapes.ttl` state). It is worth being precise: it does *not* use the closed-form law but its canonical twin — the Martin et al. (1998) power equation integrated over segments [Martin et al. 1998]. Per segment between points *a* and *b*:

- $F_{\text{roll}} = C_{\text{rr}} \cdot m \cdot g \cdot \cos\theta, F_{\text{grav}} = m \cdot g \cdot \sin\theta, F_{\text{aero}} = \frac{1}{2} \cdot \rho \cdot \text{CdA} \cdot v_{\text{eff}} \cdot |v_{\text{eff}}|$ (with $v_{\text{eff}} = v + \text{headwind}$);
- $P_{\text{wheel}} = \left(F_{\text{roll}} + F_{\text{grav}} + F_{\text{aero}}\right) \cdot v, P_{\text{pedal}} = P_{\text{wheel}}/\eta$;
- **negative power is zeroed** — $E = \max\left(P_{\text{wheel}}, 0\right)/\eta \cdot \Delta t$ — so descending and braking recover nothing. This is the per-segment analogue of the descent clamp, realised by clamping wheel power rather than through the $\epsilon \cdot \beta \cdot h_-$ recovery term.

The total is $E_{\text{total}} = \Sigma \max\left(P_{\text{wheel}}, 0\right)/\eta \cdot \Delta t$, with kcal $= \left(\text{kJ}/0.24\right)/4.184$. Speed comes from GPX timestamps when present, else from constant-effort-per-terrain target powers solved by bisection and brake-capped on descents. It shares the same physical constants (C_{rr}, `CdA`, ρ, mass, η, headwind) and the same per-regime climb/flat/descent power structure as the `canonical()` engine of §3 — making it the field deployment of the *forward* engine, complementing sampasimu's deployment of the *closed-form* one. **Deployment:** a static app (open `index.html`), with elevation from the collective's FABDEM tile server.

9.4 Summary

Project	Model used	Form of the shared l
sampasimu	Closed-form (approximate) law as a per-edge Dijkstra cost	`v2Edge`: uphill aRoll ·
amora	None — records the kJ output	`ph:energyEstimate` /
quilojaules	Canonical forward-dynamics [Martin et al. 1998], not the closed form	$\Sigma \max\left(P_{\text{wheel}}, 0\right)/\eta \cdot$

10. Discussion

10.1 What is genuinely novel

Two contributions survive every expansion of our literature search and constitute the substance of this paper; everything else is either standard physics or a clean re-packaging of it.

(i) The lumped, route-level closed-form recovery factor ε. The skeleton $E \approx \alpha \cdot x + \beta \cdot h_+$ is the textbook steady-speed energy integral and we claim nothing for it. What we have not located in the nearest prior art is a *single scalar* $\epsilon \in [0,1]$, defined at route level, that lumps all descent-specific losses (excess aerodynamic drag above the flat reference speed v_f, plus braking) into the closed form $E \approx \alpha \cdot x + \beta \cdot \left(h_+ - \epsilon \cdot h_-\right)$, together with a *geometry-only* closed form for ε itself,

$$\epsilon_{coast}(s) = \min\left(1, \alpha/(\beta s)\right), \qquad \alpha/\beta = C_{rr} + \tfrac{1}{2}\rho C_d A\left(v_f + w\right)^2/(mg),$$

drop-weighted over a profile and corrected by a near-constant calibration offset, $\epsilon \approx \text{clamp}_{[0,1]}\left(\epsilon_{\text{coast}} - 0.13\right)$. The closest cousins are all per-instant or per-speed-range constructs, never a route-level scalar credit: Bigazzi & Lindsey's negative-grade tractive-power-zero boundary [Bigazzi & Lindsey 2019] is a per-grade steady-state speed choice, not a $\beta \cdot h\_$ credit; the EV/e-bike and operations-research literature treats descent recovery as a per-instant regeneration efficiency [Yuan et al. 2024], a separate *symmetric* potential $(M+m)g \cdot \Delta H$ [Ahmadi et al. 2024], or per-edge negative arc costs solved numerically [Perger & Auer 2020] — never a calibrated $\epsilon < 1$ folded into a closed form. The −0.13 offset is the one empirical knob: it absorbs the residual descent pedalling/braking that the coasting ideal omits. On the 44 power rides it is fit in-sample and turns the $\bar{s} \geq 3\%$ median ε_coast of 0.39 into 0.26 against a measured 0.27 (worked example in §8.3); applied frozen to the censo set, it ties the flat constant selected in-sample there (RMS

0.08 vs 0.08, §8.5) — the calibration transfers across riding regimes, though the geometric ε itself over-credits on stop-go terrain (§8.4).

(ii) The energy↔time duality $x^* = x + k_+ \cdot h_+ - k_- \cdot h_-$. Neither half of the effective-flat-distance time model is itself new. The ascent half $x + k_+ \cdot h_+$ is the cycling equivalent-flat-distance idea of [Scarf & Grehan 2005] (1 m climb ≈ 8 m flat), [Scarf 2007], and [Norman 2004]; the descent half k_- is the route-level descent time-credit of [Langmuir 1984] (−10 min/300 m on gentle 5–12° descents, +10 min/300 m on steep > 12°) and [Tobler 1993] ($V = 6 \cdot e^{-3.5|S + 0.05|}$, speed peaking at −2.86°). What we have not located anywhere is the *linkage*: that k_- is the time-twin of ε, derivable from it through the single hidden descent speed v_{desc} once the descent power $\bar{P}_{\mathrm{desc}}$ is known,

$$\frac{v_f}{1 - k\_s} = \frac{\bar{P}_{desc}}{\alpha - \epsilon \beta s} \;\; \Rightarrow \;\; k_- = \frac{1}{s}\left[1 - \frac{v_f}{\bar{P}_{desc}}\left(\alpha - \epsilon \beta s\right)\right],$$

with the clean degenerate limit of a pure coast ($\bar{P}_{\mathrm{desc}} = 0$): the bridge forces $\epsilon = \alpha / (\beta \cdot s)$, pinned by grade alone, while k_- is set entirely by the terminal coasting speed — so without power to bridge them the two parameters decouple, ε purely geometric, k_- purely aerodynamic. Langmuir and Tobler are empirical time fits never tied to an energy budget; deriving the descent time-credit from the *same* descent power as the recovery factor is the genuinely additive piece. Its empirical status is settled in §8.8 and worth restating precisely: the bridge's one quantitative prediction (descent speed) fails — descents are behaviour-limited — while the ascent half transfers and the degenerate limit independently re-derives ε_coast. The duality's surviving value is structural, and we claim it as such.

10.2 What is standard, and what is additive framing

We are explicit about provenance to keep the novelty claims honest.

Component	Status
`canonical()` forward simulation	**Standard** — it *is* th
$\alpha \cdot x + \beta \cdot h_+$ energy skeleton	**Standard** (textbool
Lumped route-level ε + its closed form	**Novel** (no located p
Shared-constants closed-form-vs-sim comparison	**Incremental** framir
k_{smooth} deadband for fractal ascent *inside a closed-form law*	**Additive formaliza**
$x^* = x + k_+ \cdot h_+$ ascent half	**Standard** (equivale
k_- descent time-credit	**Has precedent**
ε↔k₋ duality (the linkage)	**Novel**
ε / k_DEM inference (invert an energy identity)	**Incremental**, Chun
São Paulo ε negative result	**Additive** (negative
Scale/terrain dependence of the behavioural constants; the calibrated deployment goal	**Additive** (empirical

The k_{smooth} story is worth a note as additive formalization rather than discovery. That cumulative ascent is fractal and scale-dependent is established [Rapaport 2011], and Rapaport even states the roller-momentum intuition in words; what we add is folding it *inside* the closed-form energy law as the totals-only scalar $k_{\text{smooth}} = 1 - c \cdot x/h_+$ with $c \approx 0.003$ (3 m/km, a ~0.3% dimensionless "noise grade"; measured 3.2 m/km, IQR 2.7–3.8). It is only needed for the approximate model — the canonical simulation tracks kinetic energy and already pays the rollers' momentum correctly. The empirical justification for $k_h = 1$ is the same sustained-climb finding of §8.2: over 2535 sustained sections measured $\int P \cdot \text{dt}$ on climbs (41 790 kJ) equals expected gravity+rolling+aero (43 333 kJ) to within 4% (ratio 0.96, $k_h(\text{sustained}) = 0.96$), so $\beta \cdot h_+$ is correct on real climbing and the over-count is entirely sub-metre noise and short rollers.

10.3 Corpus-bounded honesty

Every "novel" call in this paper means **"no precedent located in the nearest road-cycling-power and elevation-routing prior art,"** *not* "provably first." The search comprised two harness passes: a first over the cycling-power corpus and a follow-up extending into EV/e-bike energy and regeneration, electric-vehicle routing in operations research, and hiking equivalent-flat-distance time models. It did *not* sweep the full transportation, operations-research, or sports-physiology literatures. The two claims that survived *both* passes — the lumped route-level closed-form ε, and the ε↔k₋ duality — are the ones we put weight on; the lumped-ε claim was in fact *strengthened* by the EV/e-bike expansion, since that literature's descent recovery is invariably a per-instant or per-speed-range regeneration efficiency rather than a route-level scalar. We also note where a tempting analogy fails: ε is the cyclist's gravity-and-brake energy budget, not the physiological concentric/eccentric muscle efficiency of [Minetti et al. 2002], which we cite only as a conceptual asymmetry analogy.

10.4 Limitations

The limitations, in rough order of how much they bound the claims:

Conditional on measured power, not blind prediction. Both engines are *conditioned on the very ride they predict*: the canonical simulation is fed that ride's own FIT-extracted climb/flat/descent powers, and the closed form's reference speed is $v_f = \text{flatEqSpeed}(P_{\text{flat}})$ from the same ride. The accuracy figures therefore measure the *consistency of the energy accounting given the measured power*, not blind planning-mode prediction — no unconditional/planning-only figure is computed here. The 3.6% best-variant on the longões set additionally uses the rider's *hand-entered* per-ride ε from the source spreadsheet, not the closed-form ε; it is a best-of-≈9-variants headline. The nearest step toward planning mode is §8.9's calibration result: constants fitted on half a rider's history hold on the held-out half — so the constants are learnable in advance — but each held-out prediction still uses that ride's own extracted flat power, so even the goal-passing configuration is power-conditioned; a true planning-mode benchmark (power guessed before the ride) remains uncomputed.

Three riders, three power meters — but geography and rider behaviour are the open limits. The benchmarks come from three riders' own devices: the author (Datasets 1–2, 106 rides) and two independent riders, P. Paz and JAAM (Datasets 4–5, 441 + 219 rides), whose data arrived after the calibrations were frozen. Geography, however, is coincidentally rather than structurally controlled: all three riders' power/descent benchmarks fall in the São Paulo altitude band (~730 m) — JAAM's multi-country history is almost entirely in *non-power* activities (its power + real-descent non-SP subset is n = 2) — so different climates, air densities, traffic regimes, and bike categories remain untested. And the ε evaluation shares its method (30 m cells, measured flat speed) across riders, so a method-level artifact would not be caught. The rider-behaviour limit itself — the geometric ε skill working only for riders who coast — is a *finding*, stated with its evidence in §8.6.

The assumed rider physics is corroborated — and it bounds the ε margin. Datasets 4–5 assume CdA and C_rr (only mass is data-implied). An independent per-activity power-balance fit (virtual-elevation with a GPS-bearing wind vector; journal Entry 15) recovers per-rider CdA, C_rr, mass, and wind, all inside plausible ranges (P. Paz CdA ≈ 0.26 / C_rr ≈ 0.005; JAAM ≈ 0.32 / 0.011), anchor-validating on the author (recovered CdA 0.33 vs 0.39 assumed, C_rr 0.008 vs 0.008); JAAM's $\hat{m}$ = 101.7 kg, once suspected an artifact, is rider-confirmed at ≈ 100 ± 7 kg. This is a plausibility check, not a re-calibration (only ~25% of rides fit the single-rider balance cleanly; wind magnitude leans on a per-rider de-bias), and the scoreboard keeps the assumed constants. Its one substantive consequence is already in §8.6: rerunning with P. Paz's fitted constants leaves the energy accuracy intact but collapses the frozen-ε 35% margin to a tie (journal Entry 16).

The ε correlations are in-sample and part–whole. The −0.13 offset is calibrated on the same 44 rides its correlations are reported on, and ϵ_{coast} shares its dominant term and per-ride α with the "truth" ϵ_{bal}, so the headline 0.77/0.82 correlations are inflated by construction (mechanics in §8.3); the RMS-reduction figures are the defensible statistics. The transfer ledger is then short: the offset transfers across regimes (§8.5) and riders (§8.6); the energy law transfers everywhere; the geometric ε *skill* transfers nowhere robustly — it under-performs a flat constant on stop-go terrain (§8.4), ties under fitted physics for one independent rider, and is inconclusive for the other (§8.6).

The time model is only half-confirmed. The energy↔time duality $x^* = x + k_+ \cdot h_+ - k_- \cdot h_-$ is now tested against measured moving time on all three datasets (§8.8), with a split verdict: the **ascent term is empirically supported and transfers** across riders (6.6% median vs 7.6% naive at the pre-declared endpoint; significant but modest, hilly-concentrated), while the **descent ε↔k₋ bridge is not confirmed** — it over-predicts measured descent speed (correlation ≤ 0.14 on the second rider) because real descents are behaviour/cap-limited, so k_- stays a free, corpus-dependent coefficient rather than one derived from ε. The clean out-of-sample mode is power-conditioned v_f; the speed-anchored variant and the per-ride descent-speed diagnostics reuse measured time and are in-sample.

ε is behavioural, and its braking hypothesis failed. ε is not a property of terrain alone; it depends on how the rider descends (the gentle-terrain reversal, with its worked example, is in §8.3). On the urban corpus we tested the natural hypothesis that the residual ε over-credit tracks braking density and **refuted it in our own data** — no stop-go predictor clears R² ≈ 0.14, and the mechanistic correction over-corrects to worse-than-a-constant (§8.5). The practical consequence is the corpus-bounded rule already stated there: ϵ_{geom} on open coastable routes, flat $\epsilon \approx 0.20$ on urban stop-go.

The ~4–7% accuracy is self-reported against a single benchmark. All three models reproduce measured $\int P \cdot \mathrm{dt}$ to ~4–7% median error with a generic assumed rider — on the 62 clean censo rides the poor-man's scalar k_{smooth} at ε = 0.20 reaches 3.9% median |Δ%|, as accurate as the forward simulation at 6.5% — but this is the repo's own self-report scored against one ground truth, the power-meter $\int P \cdot \mathrm{dt}$ integral. There is no *external*, independently measured integrated-energy benchmark; the prior art validates instantaneous speed/power on controlled tracks ([Martin et al. 1998] on a flat taxiway; [Dahmen & Saupe 2011] on rural-track speed, excluding steep descents and braking), and the nearest real-non-racing-data precedent [Gebhard et al. 2016] predicts battery range, not mechanical $\int P \cdot \mathrm{dt}$. The accuracy figures should be read as *internal consistency against the power meter*, not as validation against an orthogonal energy measurement.

Parameter/benchmark caveats. The censo rider physics is *assumed* (m = 78 kg, C_dA = 0.40, C_rr = 0.008, ρ = 1.13, wind = 0, k_eff = 0.98), and the descent-balance $\epsilon \approx 0.23$ may still be a touch deflated by $C_{\mathrm{rr}} = 0.008$ being low for rough city asphalt. The physical-floor filter (legE ≥ $\mathrm{mg} \cdot h_+/k_{\mathrm{eff}}$) and cadence check excluded 7 rides that measured below the climbing potential energy (down to 53%, over-predicting by +79…+373%) — a data-quality cut that is principled but does prune the corpus.

10.5 An alternative considered: regime decomposition

A structurally cleaner alternative was tested and rejected (journal Entry 17): decompose the ride by slope regime and let each regime evaluate the base law with its **own** power and modelled speed — $E_{\mathrm{new}} = E_{\mathrm{flat}}(x_=; P_=) + E_{\mathrm{climb}}(x_+; P_+) + E_{\mathrm{descent}}(x_-; P_-)$ (climb aero at the quasi-steady $v_c(P_+)$; three descent treatments incl. a no-ε $P_- \cdot t_-$), plus a totals variant $\alpha(P_=) \cdot x + \beta \cdot h_+ - \epsilon \cdot \beta \cdot h_-$. Evaluated on regime totals — matching how the champion itself evaluates; a per-edge realisation with a *ride-frozen* ε discards ε's aggregate physicality and over-charges descents, while the deployed grade-local variant (§9.1) avoids that but is resolution-sensitive (journal Entry 18) — the decomposition is competitive but **not an improvement**: it loses the pre-declared second-rider endpoint (best variant 6.2% vs the champion's 5.8%), ties on the unbiased author corpus, and wins only where the champion under-predicts. A fitted-physics rerun makes the mechanism causal (6 of 6 corpus×physics configurations): its wins are a **bias trade** — a near-constant ~+4.6 pp climb-aero padding that helps exactly when the parameter set under-predicts — not a structural gain, even though it consumes all three regime powers where the champion's closed form needs only P_{flat}. The champion's simplifications (zero climb aero, lumped ε) earn their keep as bias-cancellation. One transferable idea survives: the flat-resistance grade α/β is the natural *scale* of the regime threshold (1.4–2.5% across corpora, ordering with rider speed, sitting at the 2% default), though a symmetric ±α/β rule underperforms the asymmetric per-corpus optimum.

11. Conclusion and future work

We presented two engines for the mechanical energy of pedalling a route — a standard Martin-1998 forward simulation [Martin et al. 1998] and the cheap closed form $E \approx \alpha_r \cdot x + \alpha_a \cdot x_{\mathrm{flat}} + k_h \cdot k_{\mathrm{smooth}} \cdot \beta \cdot (h_+ - \epsilon \cdot h_-)$ — run on *shared* physical constants, so that the gap between them isolates modelling error from parameter error.

The most consequential empirical finding is attributional: the closed form's error is not diffuse but two identifiable artifacts — the climb-aero over-charge and fractal ascent noise — and correcting them (the α-split; a 2 m deadband, or $k_{\mathrm{smooth}} = 1 - c \cdot x/h_+$ with $c \approx 3m/\mathrm{km}$) takes it from 19.3% to **3.6% median error — statistical parity with the forward simulation itself** (5.1%; the paired per-ride difference is not significant, §8.1). On the descent side, the lumped recovery factor ε, with its geometry-only estimator $\epsilon \approx \mathrm{clamp}_{[0,1]}(\min(1, \alpha/(\beta \cdot \bar{s})) - 0.13)$, delivered a split verdict that took six datasets to earn: descent recovery is unambiguously real ($\epsilon = 0$ over-predicts every corpus), and the calibrated −0.13 offset recurs on every rider tested (gaps 0.12–0.13) — but the geometric *skill* beyond a flat constant is fragile, winning for a coasting rider only under the generic assumed physics and tying under his own fitted constants, inconclusive for a descent-pedaller (§8.3, §8.5, §8.6). The ε↔k₋ **energy↔time duality** $x^* = x + k_+ \cdot h_+ - k_- \cdot h_-$ fared the same way its energy sibling did: the ascent half transfers to an unseen rider's measured times, the descent bridge does not, and the duality's surviving value is structural (§8.8). We also reported a per-source DEM correction k_{DEM} = IGC/source (FABDEM 0.95, COP30 0.84, SRTM 0.75 against 5 m bare-earth survey truth).

Deployment closed the loop with a finding of its own (§8.9): the law's *behavioural* constants (k_{smooth}, the ε offset, the climb threshold) are functions of the elevation-sampling interval and terrain regime, not universals — the frozen law over-charges on the deployed 5 m survey DEM, a rider-independent re-fit of the trio bridges the gap only within the terrain regime it was fitted on, and with a σ = 10 m raster pre-smoothing plus per-rider constants calibrated on each rider's own history the pipeline meets a pre-registered ±5% error / ±2% bias goal on held-out rides for all three riders. Within its family, the closed form is already good enough; the constants are the accuracy frontier, and they are learnable from history.

The law is deployed at three fidelities across the collective's software: verbatim as a per-edge Dijkstra cost over DEMs in *sampasimu* (which also ships the σ = 10 m resolution mitigation), as a recorded per-ride `ph:energyEstimate` kJ literal in *amora*, and alongside the canonical forward engine in *quilojaules*.

Two open questions follow directly from the limitations and bound any claim of generality.

1. An apples-to-apples integrated-energy benchmark. Our ~4–7% median error is measured only against the power meter's own $\int P \cdot \mathrm{dt}$. We have no *independent* integrated-energy reference, and the controlled-track prior art validates instantaneous quantities, not whole-route energy. Closing this gap means a benchmark that measures total mechanical energy by a path orthogonal to the power channel — a controlled-rider, instrumented protocol, or cross-validation against a measurement that does not itself integrate the same power stream — so the ~4–7% can be attributed to the model rather than to internal consistency with the benchmark it is scored against.

2. Does k_{smooth}-in-an-energy-law have a GPS-track-energy precedent? That cumulative ascent is fractal and scale-dependent is established [Rapaport 2011], and equivalent-flat-distance time models exist for ascent [Scarf & Grehan 2005; Scarf 2007]; but we did not locate prior work that folds a scale-dependent ascent correction *into a closed-form energy law* over real GPS/DEM tracks, as our totals-only $k_{\text{smooth}} = 1 - c \cdot x/h_+$ does. Whether such a precedent exists in the broader transportation or geomorphometry literature — outside the corpus we swept — is an open question that would settle whether this piece is a genuine formalization or a re-derivation. Resolving it, along with a behaviour-aware model of ε that the São Paulo negative result shows is needed (ε depends on *how* one descends, not only the grade), is the natural next step — as is a terrain-roughness-indexed form of the behavioural constant trio, which §8.9 shows a single terrain-blind constant cannot span.

Data and code availability

All code is public: the comparison app (`energy-model-comparison.html`), the validation harnesses under `data/activities/` (`compare`, `censo_compare`, `eps_hypothesis`, `eps_sp_test`, `time_compare`, the per-rider inventory + compare pairs `ppaz_*` / `jaam_*` / `danlessa_*`, the parameter-estimation `param_fit` / `cda_estimate`, the regime-decomposition test `regime_compare`, the resolution/deployment studies `verify_v2edge_clamp` /

`igc_resolution_test` / `goal_calibration` / `scale_trio`, and the CI/paired-test computation `bootstrap_ci`), and the derivations (`notas.md`) live at https://github.com/danlessa/bicycling-energy-model; the deployed sibling tools (sampasimu, amora, quilojaules) are under the `pedalhidro` GitHub organisation. The raw ride recordings (FIT files carrying GPS tracks) and the source spreadsheets are **not** published — they contain location and private-activity data — but every number in this paper regenerates from them with one command per harness, and aggregated per-ride CSVs with coordinates stripped are available on request. The analysis provenance is logged entry-by-entry, with commit hashes, in `research/MODEL_COMPARISON_JOURNAL.md`.

Ethics and privacy

The power and GPS recordings analysed here come from three riders: the author (Datasets 1–2, on the author's own device, including on group rides) and **two independent riders — not members of the collective — who each shared their full activity-history export with informed consent** for this analysis (Datasets 4–5, P. Paz and JAAM; held locally, never published, referred to only by initials). No other rider's data is used — censo activities recorded by third parties (16 links) were excluded rather than accessed. Aggregates reported for the two shared datasets (energies, ε, implied mass, and non-locational altitude/terrain summaries) carry no coordinate information. The repository history was audited for leaked personal data; all raw tracks, exports, and spreadsheets are excluded from version control.

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To the *Pedal Hidrográfico* collective — the rides, the census, and the reason this model exists. *Seguir as águas*.

References

- **[Ahmadi et al. 2024]** Ahmadi, S., Tack, G., Harabor, D., Kilby, P. & Jalili, M. (2024). *Efficient Energy-Optimal Path Planning for Electric Vehicles Considering Vehicle Dynamics*. arXiv:2411.12964 (v1 Nov 2024; rev. Mar 2026). https://arxiv.org/abs/2411.12964
- **[Bigazzi & Lindsey 2019]** Bigazzi, A. & Lindsey, R. (2019). *A utility-based bicycle speed choice model with time and energy factors*. Transportation 46(3):995–1009. https://doi.org/10.1007/s11116-018-9907-2
- **[Chung]** Chung, R. *Estimating CdA with a power meter* (the "virtual elevation" method). http://anonymous.coward.free.fr/wattage/cda/indirect-cda.pdf
- **[Dahmen & Saupe 2011]** Dahmen, T. & Saupe, D. (2011). *Validation of a Model and a Simulator for Road Cycling on Real Tracks*. https://www.uni-konstanz.de/mmsp/pubsys/publishedFiles/DaSa11.pdf
- **[Danek et al. 2020]** Danek, T., Slawinski, M. A. & Stanoev, T. (2020). *On modelling bicycle power-meter measurements: Part I. Estimating effects of air, rolling and drivetrain resistance*. arXiv:2005.04229. https://arxiv.org/abs/2005.04229

- **[di Prampero et al. 1979]** di Prampero, P. E., Cortili, G., Mognoni, P. & Saibene, F. (1979). *Equation of motion of a cyclist*. J. Appl. Physiol. 47(1):201–206. PMID 468661.
- **[Gebhard et al. 2016]** Gebhard, V. et al. (2016). *Range prediction for electric bicycles* (WeBike, OSM). Proc. 7th ACM Int. Conf. on Future Energy Systems (e-Energy). https://doi.org/10.1145/2934328.2934349
- **[Langmuir 1984]** Langmuir, E. (1984). *Mountaincraft and Leadership*. Scottish Sports Council / MLTB. (Descent corrections to Naismith's rule.)
- **[Li et al. 2025]** Li, X., Zou, B., Wang, X. & Liu, C. (2025). *Optimization of physical energy and velocity allocation for cyclists in road cycling individual time trial using genetic algorithm*. Front. Physiol. 16:1683815. https://doi.org/10.3389/fphys.2025.1683815
- **[Martin et al. 1998]** Martin, J. C., Milliken, D. L., Cobb, J. E., McFadden, K. L. & Coggan, A. R. (1998). *Validation of a Mathematical Model for Road Cycling Power*. J. Appl. Biomech. 14(3):276–291. https://doi.org/10.1123/jab.14.3.276
- **[Minetti et al. 2002]** Minetti, A. E., Moia, C., Roi, G. S., Susta, D. & Ferretti, G. (2002). *Energy cost of walking and running at extreme uphill and downhill slopes*. J. Appl. Physiol. 93(3):1039–1046. https://doi.org/10.1152/japplphysiol.01177.2001 (conceptual asymmetry analogy only.)
- **[Norman 2004]** Norman, J. M. (2004). *Running uphill: energy needs and Naismith's rule*. J. Oper. Res. Soc. 55(3):308–311.
- **[Perger & Auer 2020]** Perger, T. & Auer, H. (2020). *Energy efficient route planning for electric vehicles with special consideration of the topography and battery lifetime*. Energy Efficiency 13:1705–1726. https://doi.org/10.1007/s12053-020-09900-5
- **[Rapaport 2011]** Rapaport, D. C. (2011). *Evaluating cumulative ascent: Mountain biking meets Mandelbrot*. Int. J. Mod. Phys. C 22(3):209–217. arXiv:1011.4778. https://arxiv.org/abs/1011.4778
- **[Scarf 2007]** Scarf, P. (2007). *Route choice in mountain navigation, Naismith's rule, and the equivalence of distance and climb*. J. Sports Sci. 25(6):719–726. PMID 17454539.
- **[Scarf & Grehan 2005]** Scarf, P. & Grehan, P. (2005). *An empirical basis for route choice in cycling*. J. Sports Sci. 23(9):919–925. https://doi.org/10.1080/02640410400023282
- **[Tobler 1993]** Tobler, W. (1993). *Three presentations on geographical analysis and modeling*. NCGIA Technical Report 93-1. (Hiking function $V = 6 \cdot e^{-3.5|S + 0.05|}$.)
- **[Yuan et al. 2024]** Yuan, X., He, J., Li, Y., Liu, Y., Ma, Y., Bao, B., Gu, L., Li, L., Zhang, H., Jin, Y. & Sun, L. (2024). *Data-driven evaluation of electric vehicle energy consumption for generalizing standard testing to real-world driving*. Patterns 5(4):100950. https://doi.org/10.1016/j.patter.2024.100950
